SUSTech Southern University of Science and Technology

EE104 Fundamentals of Electric Circuits (Lesson 04)

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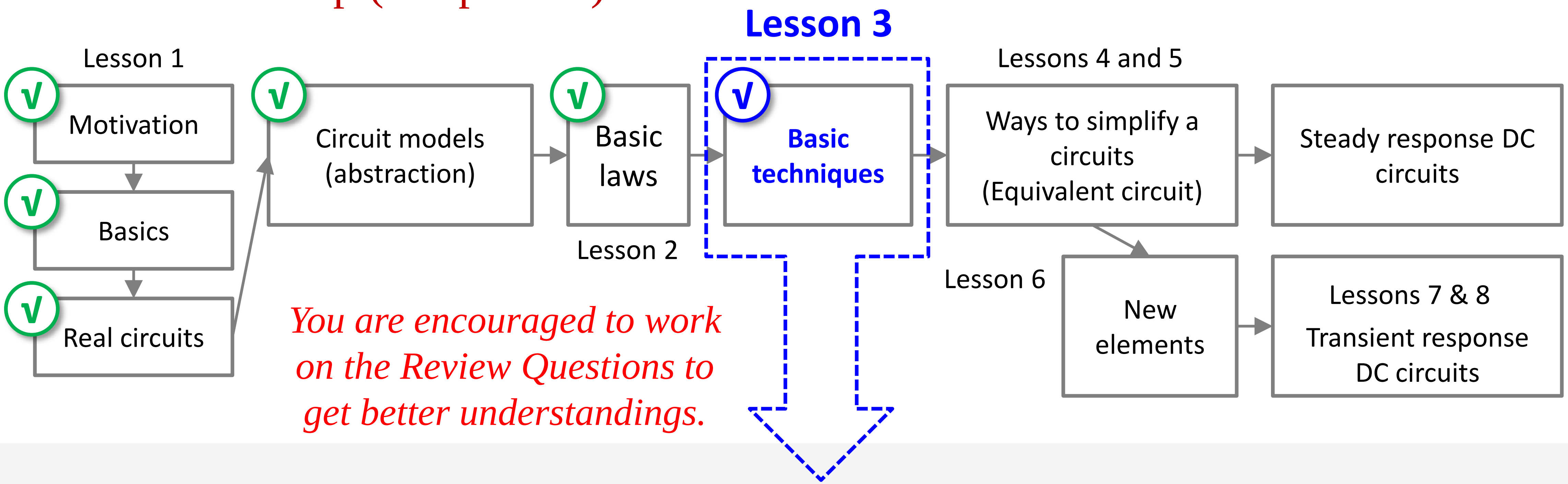
Review of Lesson 3 (a)

- EE104 roadmap (DC portion)



You are encouraged to work on the Review Questions to get better understandings.

Nodal analysis & Mesh analysis

(planar circuits & supernode & supermesh)

Review of Lesson 3 (b)

- Nodal analysis:

Step 01 **Label the Nodes**	a) Pick any node as the voltage reference. Label its voltage as 0 V. Label dependent sources with v_s or i_s. b) If any voltage sources are connected to a labelled node, label their other ends by adding the value of the source onto the voltage of the labelled end. Repeat as many times as possible. c) Pick an unlabelled node and label it, then loop back to step b) until all nodes are labelled. d) For each dependent source, write down an equation that expresses its value in terms of other node voltages.
Step 02 **KCL at nodes**	e) Write down a KCL equation for each "normal" node (i.e. one that is not connected to a floating voltage source). f) Write down a KCL equation for each "super-node". A super-node consists of a set of nodes that are joined by floating voltage sources and includes any other components joining these nodes.
Step 03 **Ohm's Law**	g) Substitute the currents using Ohm's Law
Step 04 **Solve equations**	h) Solve the equation

Review of Lesson 3 (c)

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- Mesh analysis:

Step 01 Label the Mesh	◻ Identify the mesh and assign a mesh current to each of them ◻ Select the same direction for all mesh currents (i.e. clockwise)
Step 02 KVL for meshes	◻ Apply KVL for each mesh based on reference voltage directions ◻ Each mesh results in an equation in terms of mesh currents
Step 03 Ohm's Law	◻ Apply Ohm's Law to express each voltage ◻ Pay extra attention to the reference voltage directions
Step 04 Solve equations	◻ Solve the equations obtained in previous steps. ◻ Number of equations will be equal to the number of meshes

Review of Lesson 3 (d)

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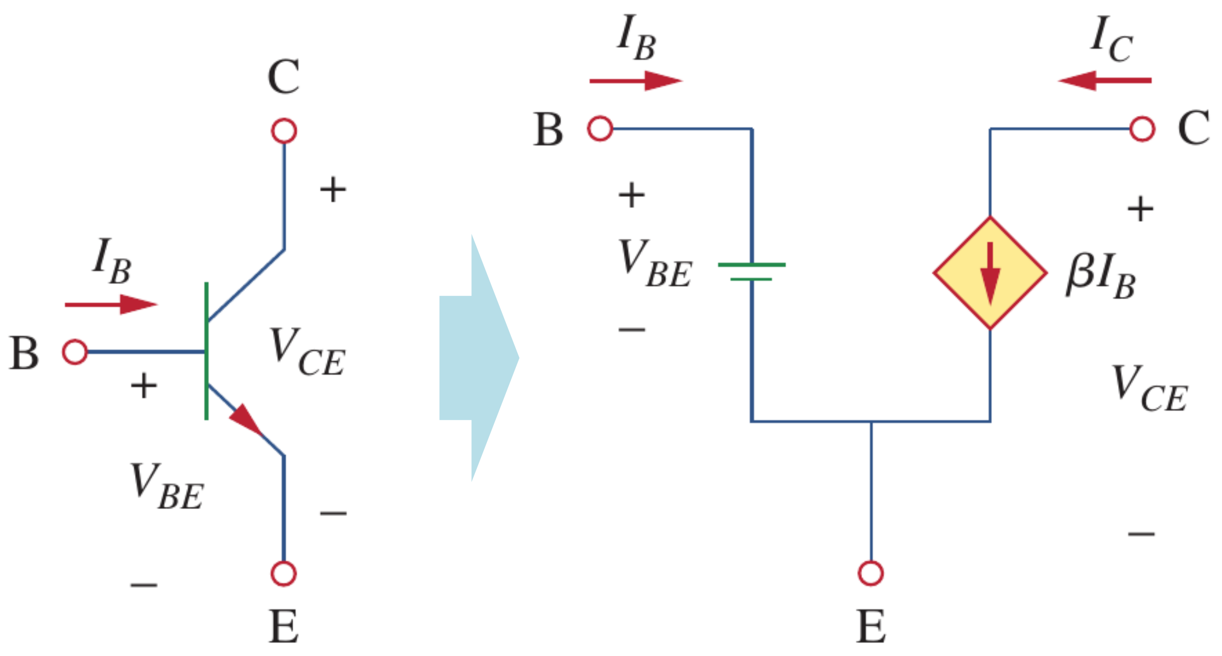
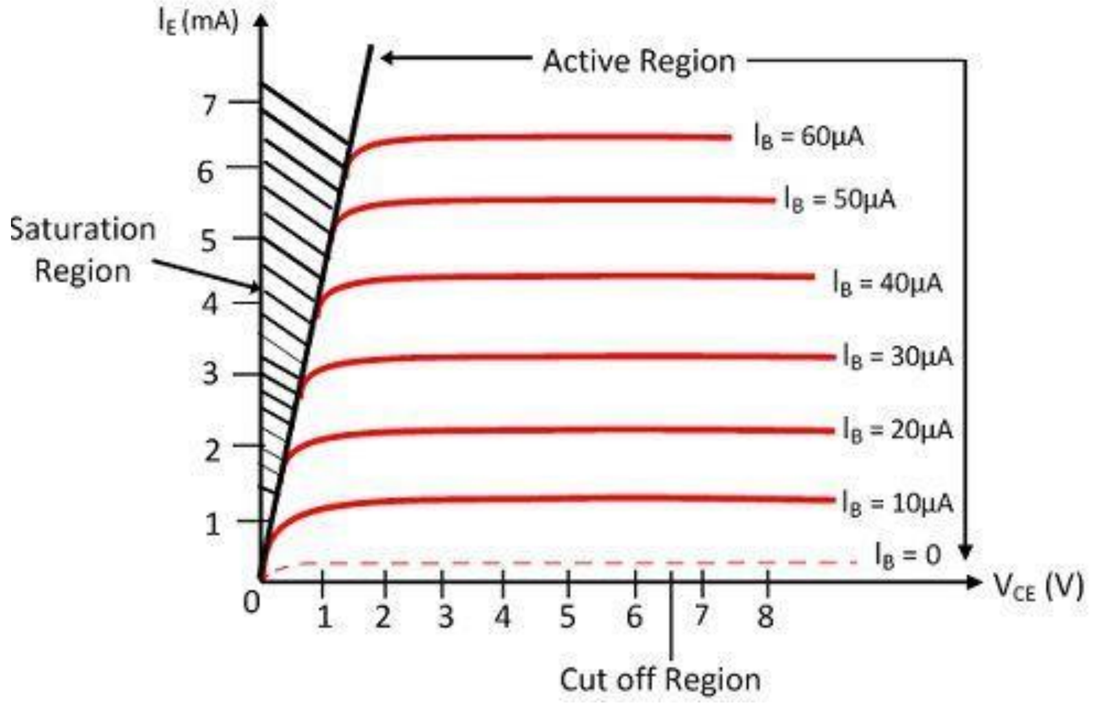
- Equivalent circuits of transistors in DC circuits:

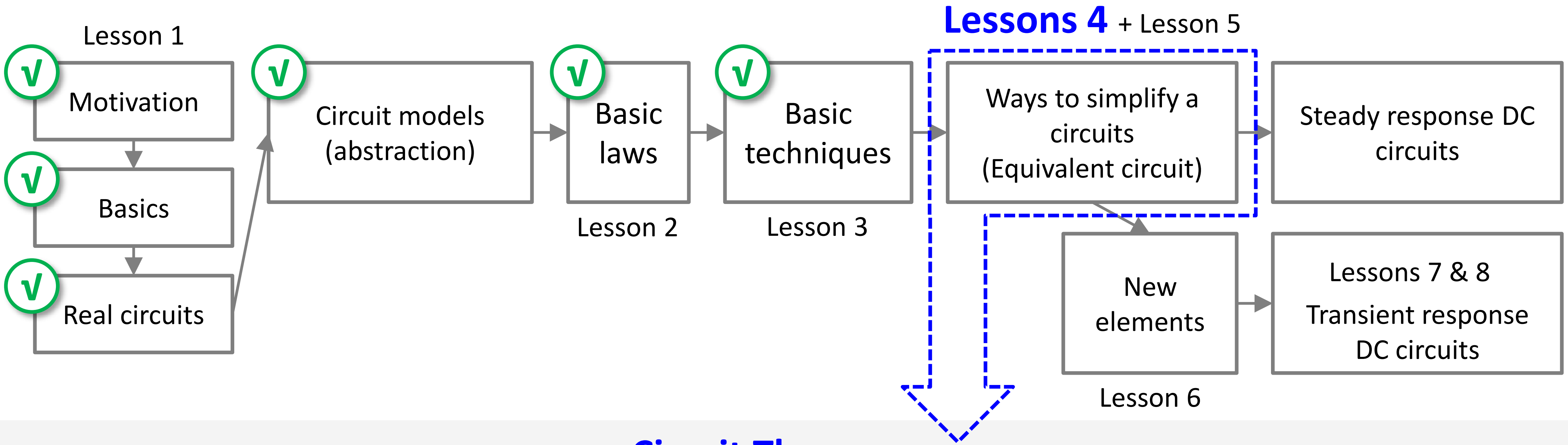




$$I_E = I_B + I_C, \quad I_E = (1+\beta)\cdot I_B$$

Review of Lesson 3 (e)

- EE104 roadmap (DC portion)



Circuit Theorems

1. Linear circuits

2. Superposition theorem

3. Source transformation theorem

4. Thévenin's theorem

5. Norton's theorem

6. Maximum power transfer theorem

Lesson 4 Circuit Theorems

4.1-4.4	**Linear circuits and superposition theorem**	**Slides 08-31**
4.5-4.6	Source transformation theorem	Slides 33-47
4.7	Thévenin’s theorem	Slides 49-60
4.8-4.9	Norton’s theorem	Slides 62-70
4.10	Maximum power transfer theorem	Slides 72-80
4.11	Summary and assignments	Slides 82-83

4.1 Linearity (a)

- Circuit theorems

Complex circuits → Circuit theorems → Simplified circuits for analysis

a) A major advantage of analyzing circuit using Kirchhoff's laws as we did before is that we can analyze a circuit without tampering with its original configuration, however the major disadvantage is that tedious computation is involved for a large and complex circuit.

b) To handle the complexity, engineers over the years have developed some theorems to simplify circuit analysis, including Thevenin's theorem and Norton's theorem.

c) Since these theorems are applicable to a linear circuit, we first discuss the concept of circuit linearity, in addition to superposition, source transformation, and maximum power transfer.

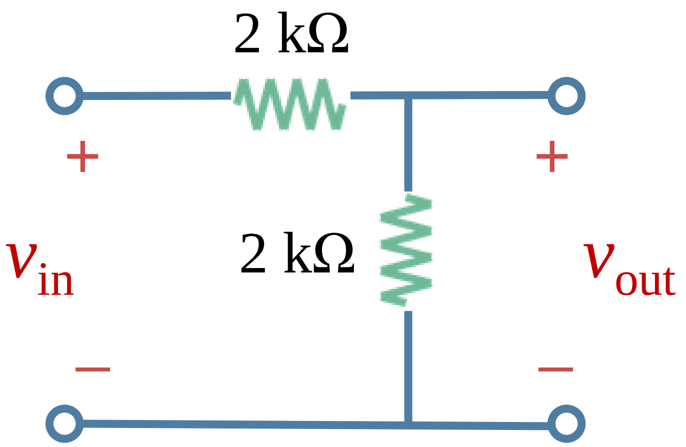
4.1 Linearity (b)

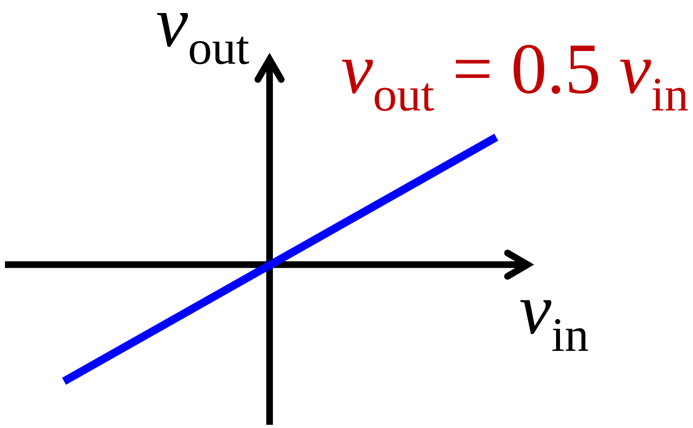
- Linearity:

a) Linearity (线性) is the property of an element describing a linear relationship between cause and effect, which is a combination of both the homogeneity (scaling) property (齐次性) and the additivity property (叠加性).

b) In a linear circuit (线性电路), the response of the output is directly proportional to the input, which means that the amplitude of the output of the linear circuit is a fraction, or a multiple of the amplitude of the input.

c) In simple words, linear circuits are networks based on linear elements (resistors), independent sources, linear independent sources, which means its parameters (resistance, inductance, capacitance, waveform, frequency etc) are not changed with *i* and *v*.





4.1 Linearity (c)

- Homogeneity and additivity

Input/ Excitation (输入/激励) $\xrightarrow{x}$ **Element/Circuit** $y = f(x)$ $\xrightarrow{y}$ Output/ Response (输出/响应)

- The homogeneity (scaling) property (齐次性): *Let* $f(x) = y,\ f(Ax) = Af(x) = Ay$

- The additivity (叠加性) property: $Let\ \begin{cases} y_1 = f(x_1) \\ y_2 = f(x_2) \end{cases},\quad We\ \ have\ \ f(x_1 + x_2) = y_1 + y_2$

- Linearity (线性): $f(Ax_1 + Bx_2) = Af(x_1) + Bf(x_2) = Ay_1 + By_2$

4.1 Linearity (d)

- Homogeneity and additivity

Input/ Excitation (输入/激励) $\xrightarrow{x(t)}$ **Element/Circuit** $y = L\{x(t)\}$ $\xrightarrow{y(t)}$ Output/ Response (输出/响应)

$$1.\ If\ y(t) = L\{x(t)\} = C\frac{dx(t)}{dt}, is\ it\ a\ linear\ element?$$

Homogeneity? Yes: $$L\{ax(t)\} = C\frac{d(ax(t))}{dt} = a \cdot C\frac{dx(t)}{dt} = a\,L\{x(t)\}$$

Additivity? Yes $$L\{x_1(t) + x_2(t)\} = C\frac{d(x_1(t) + x_2(t))}{dt} = C\frac{dx_1(t)}{dt} + C\frac{dx_2(t)}{dt}$$

$$= L\{x_{1(t)}\} + L\{x_{2(t)}\}$$

Interesting to think: are capacitors and inductors linear elements?

4.1 Linearity (e)

- Homogeneity and additivity

Input/ Excitation (输入/激励) $\xrightarrow{x(t)}$ [Element/Circuit $y = L\{x(t)\}$] $\xrightarrow{y(t)}$ Output/ Response (输出/响应)

2. $If\ y(t) = L\{x(t)\} = C, is\ it\ a\ linear\ element?$

Homogeneity? No: $L\{ax(t)\} = C \quad \neq \quad a\,L\{x(t)\} = aC$

Additivity? No $L\{x_1(t) + x_2(t)\} = C \quad \neq \quad L\{x_1(t)\} + L\{x_2(t)\} = 2C$

4.1 Linearity (f)

- Homogeneity and additivity

Input/ Excitation (输入/激励) — x(t) → Element/Circuit $y = L\{x(t)\}$ → y(t) — Output/ Response (输出/响应)

3. $If\ y(t) = L\{x(t)\} = t^2 x(t), is\ it\ a\ linear\ element?$

Homogeneity? Yes: $L\{ax(t)\} = t^2 ax(t) = a \cdot t^2 x(t) \ = a\, L\{x(t)\}$

Additivity? Yes $L\{x_1(t) + x_2(t)\} = t^2[x_1(t) + x_2(t)] = t^2 x_1(t) + t^2 x_2(t)$

$$= L\{x_{1(t)}\} + L\{x_{2(t)}\}$$

4.1 Linearity (g)

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- Homogeneity and additivity

Input/ Excitation (输入/激励) → x(t) → Device/Circuit $y = L\{x(t)\}$ → y(t) → Output/ Response (输出/响应)

$4. \mathit{If}\ y(t) = L\{x(t)\} = 3x(t) + 2, \mathit{is\ it\ a\ linear\ element?}$

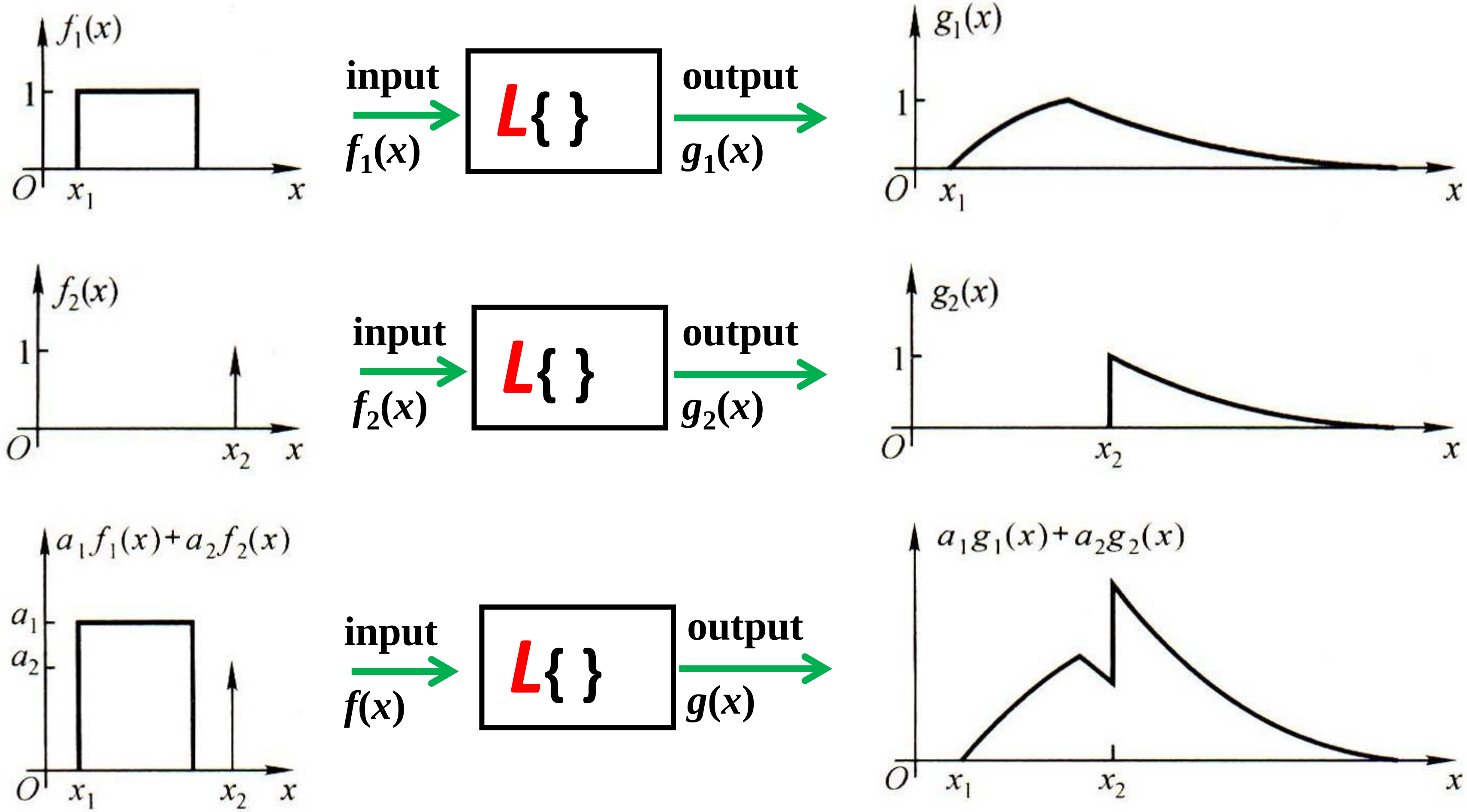
Homogeneity? No: $L\{ax(t)\} = 3ax(t) + 2 \neq aL\{x(t)\} = 3ax(t) + 2a$

Additivity? No

$$L\{x_1(t) + x_2(t)\} = 3[x_1(t) + x_2(t)] + 2 \neq L\{x_1(t)\} + L\{x_2(t)\} = 3[x_1(t) + \mathrm{x}_2(\mathrm{t})] + 4$$

4.1 Linearity (h)

- Homogeneity and additivity (time varying sources)



$$L\{f_1(x)\} = g_1(x)$$

$$L\{f_2(x)\} = g_2(x)$$

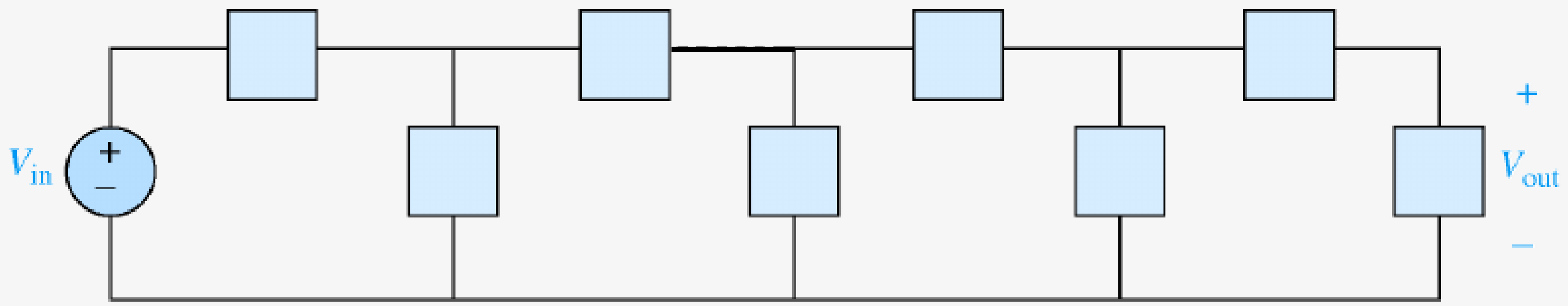
$$L\{Af_2(x)+Bf_2(x)\} = Ag_1(x) + Bg_2(x)$$

4.2 Proportionality property (a)

- Proportionality Property/Homogeneity（比例性/齐性定理）

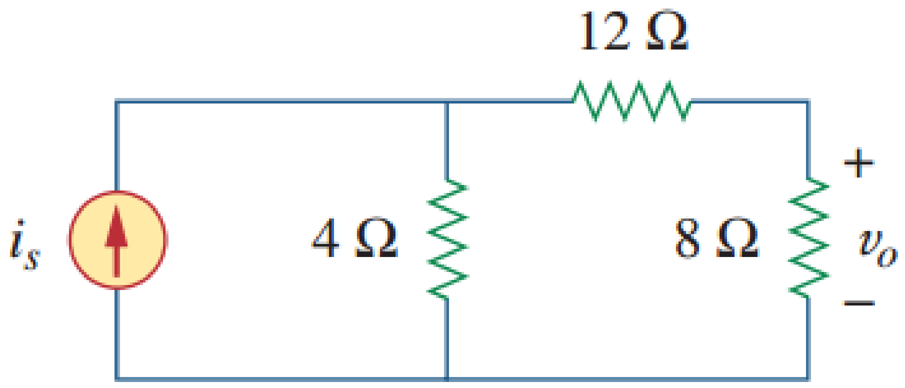
a) In a linear circuit, when only one independent source is acting (alone), the output is proportional to that input (refer to linearity theorem). Since the system is linear if the input is multiplied by a constant, then the output is multiplied by the same constant.

b) An interesting and illustrative example of usefulness of this proportionality property is the analysis of a ladder network (a circuit which has a patterned structure shown below where each box is a two-terminal element:



4.2 Proportionality property (b)

- Example 4.02



Example 4.02: in the following circuit, find the v_o when i_s = 30 A and i_s = 45 A.

Solution:

Assume $v_o = 8$ V, then $i_s = 6$ A, $v_o = f(6\text{A}) = 8$ V

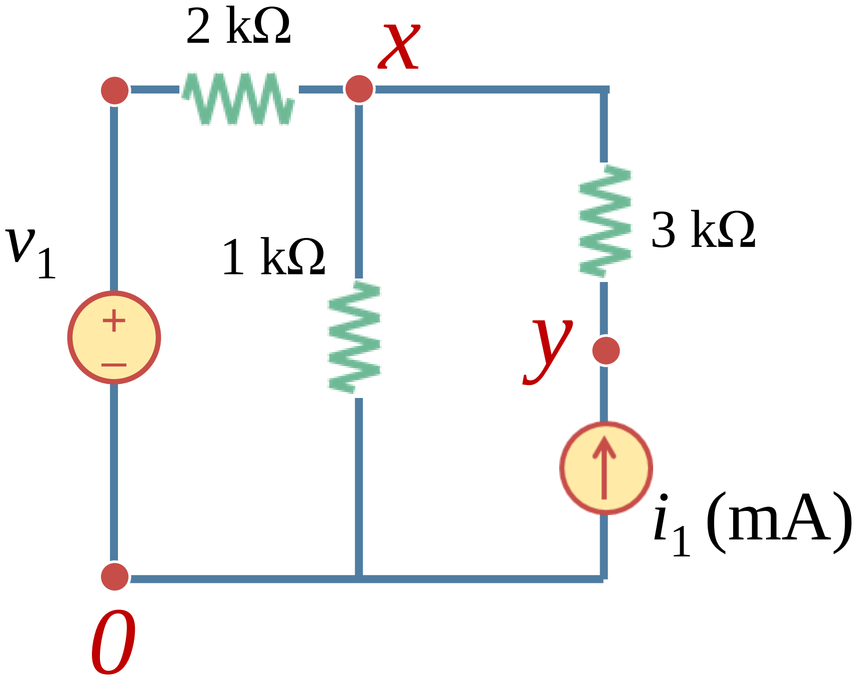
According to Proportionality Property,

When $i_s = 30$ A, $v_o = f(30\text{ A}) = f(5 \times 6\text{ A}) = 5 \times f(6\text{ A}) = 5 \times 8 = 40$ V;

When $i_s = 45$ A, $v_o = f(45\text{ A}) = f(7.5 \times 6\text{ A}) = 7.5 \times f(6\text{ A}) = 60$ V;

- Example 4.01:

Example 4.01: Prove that all node voltages in the following linear circuit can be represented in terms of the independent voltage and current sources.



Solution:

(1) Label the nodes and list KCL equations

$$-\frac{x-v_1}{2}-\frac{x}{1}+\frac{y-x}{3}=0 \quad \& \quad i_1-\frac{y-x}{3}=0$$

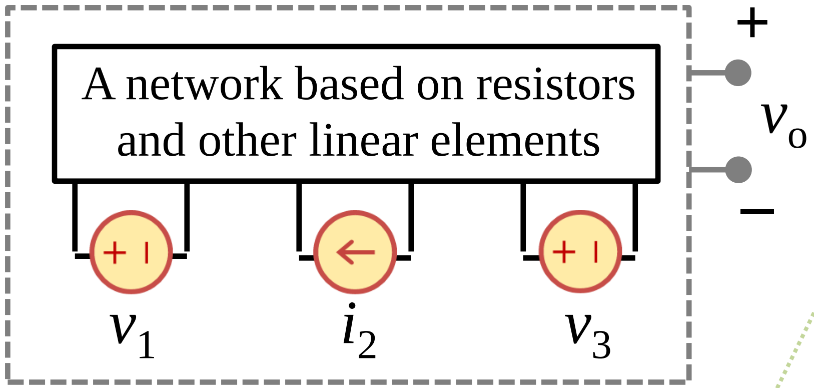
(2) Solve the nodal voltages

$$x=\frac{1}{3}v_1+\frac{2}{3}i_1 \quad \& \quad y=\frac{1}{3}v_1+\frac{11}{3}i_1$$

4.3 Superposition Theorem (c)

- Superposition Theorem

In a linear system, the superposition principle (叠加原理) states that the voltage across (or current through) an element is the algebraic sum of those caused by each independent source acting alone (每个独立源单独作用).



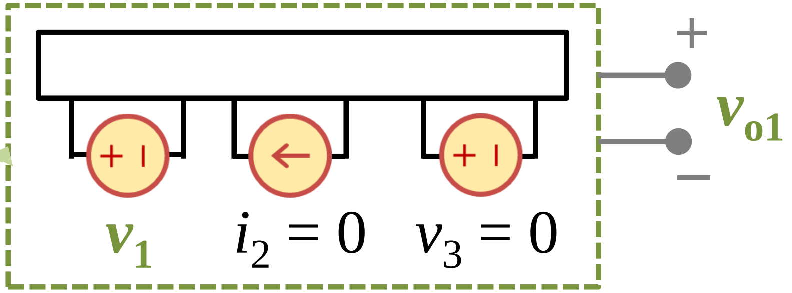
$$v_o = Av_1 + Bi_2 + Cv_3$$

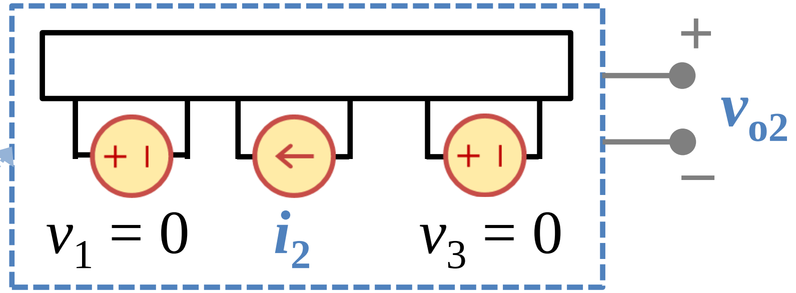
$$= \mathbf{Av_1} + B\times 0 + C\times 0$$

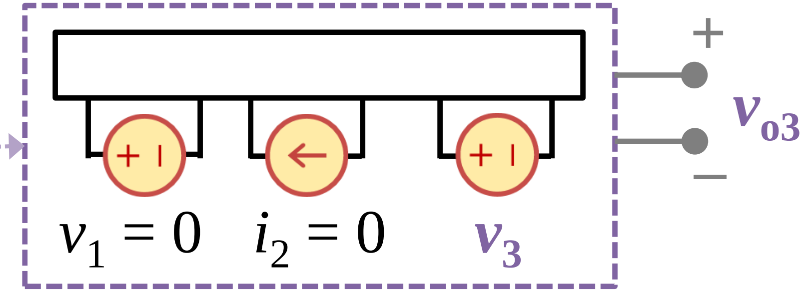
$$+ A\times 0 + \mathbf{Bi_2} + C\times 0$$

$$+ A\times 0 + B\times 0 + \mathbf{Cv_3}$$

$$= \mathbf{v_{o1} + v_{o2} + v_{o3}}$$







4.3 Superposition Theorem (d)

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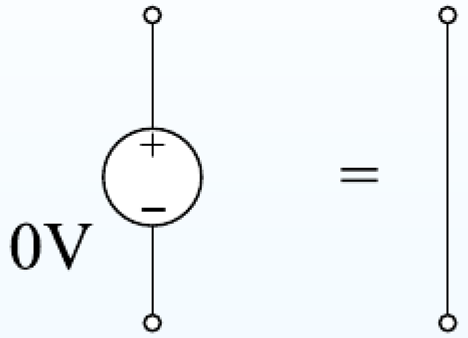
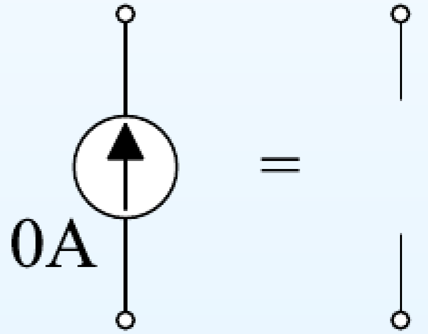
- Zero-value sources (How to turn off independent sources)



A zero-valued voltage source has zero volts between its terminals for any current. It is equivalent to a short-circuit or piece of wire or resistor of 0 Ω (or ∞ S).



A zero-valued current source has no current flowing between its terminals. It is equivalent to an open-circuit or a broken wire or a resistor of ∞ Ω (or 0 S).

4.3 Superposition Theorem (e)

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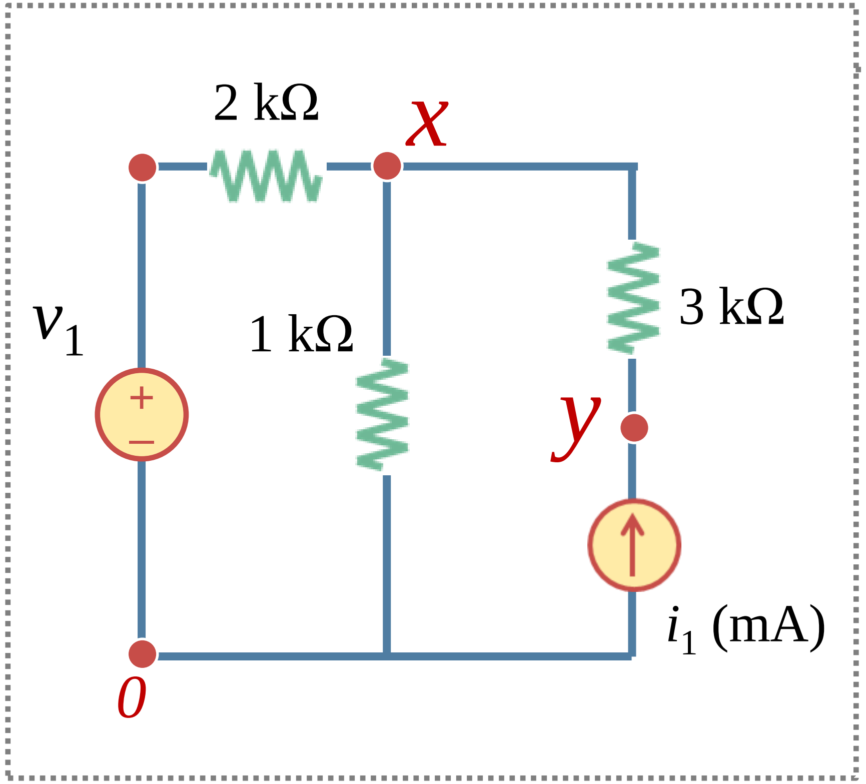
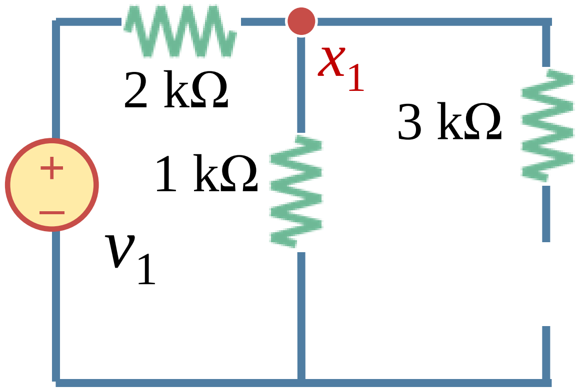
- Superposition of circuits



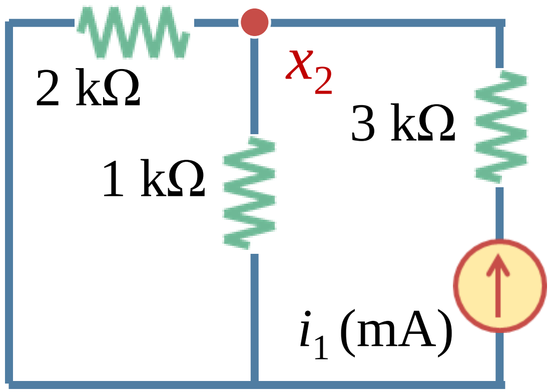
$$x = \frac{1}{3}v_1 + \frac{2}{3}i_1$$

$$x_1 = \frac{1\text{k}}{1\text{k}+2\text{k}}v_1 = \frac{1}{3}v_1$$



If only v_1 is ON

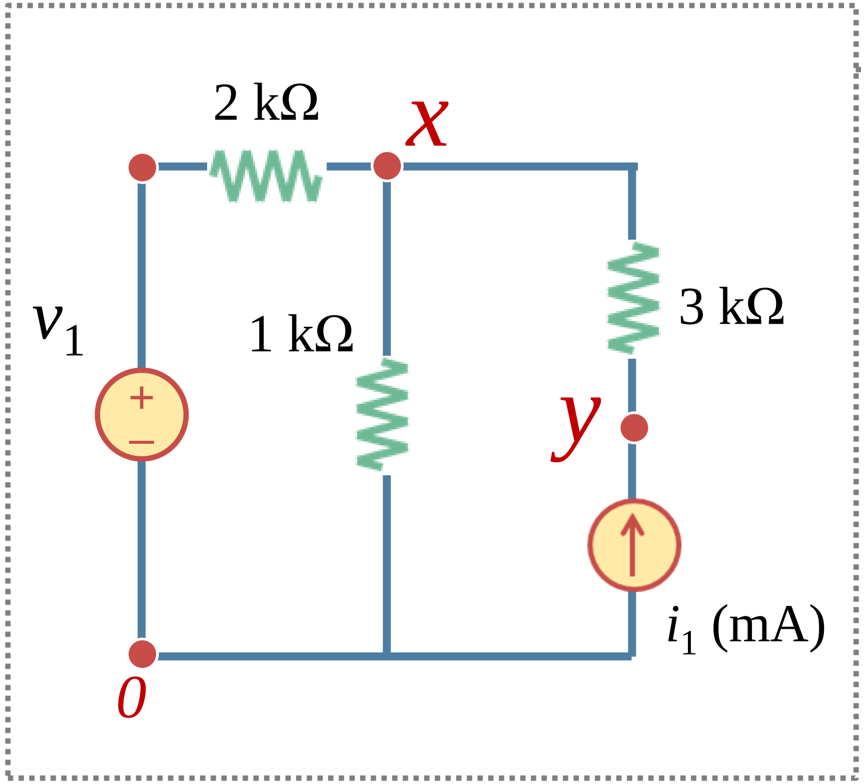
$$x_2 = \frac{2}{3}i_1$$



if only i_1 is ON

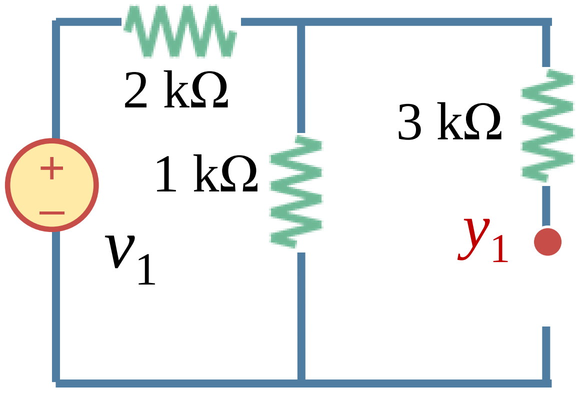
4.3 Superposition Theorem (f)

- Superposition of circuits



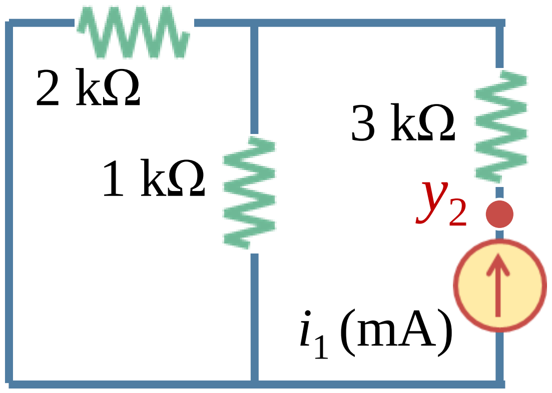
$$y = \frac{1}{3}v_1 + \frac{11}{3}i_1$$

$$y_1 = \frac{1\text{k}}{1\text{k}+2\text{k}}v_1 = \frac{1}{3}v_1$$



If only v_1 is ON

$$y_2 = (1\text{k}\|2\text{k} + 3\text{k})i_1 = \frac{11}{3}i_1$$



if only i_1 is ON

4.3 Superposition Theorem (g)

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- Steps to apply the Superposition Theorem

a) Steps to apply Superposition Theorem to analyze a circuit:

   1. Turn off all independent sources except one. Find the output;
   2. Repeat step 1 for each of all the remaining independent sources;
   3. Adding up algebraically all the obtained outputs.

b) Turn off a source means to shorten voltage sources (set to 0 V) and/or to open current sources (setting to 0 A);

4.3 Superposition Theorem (h)

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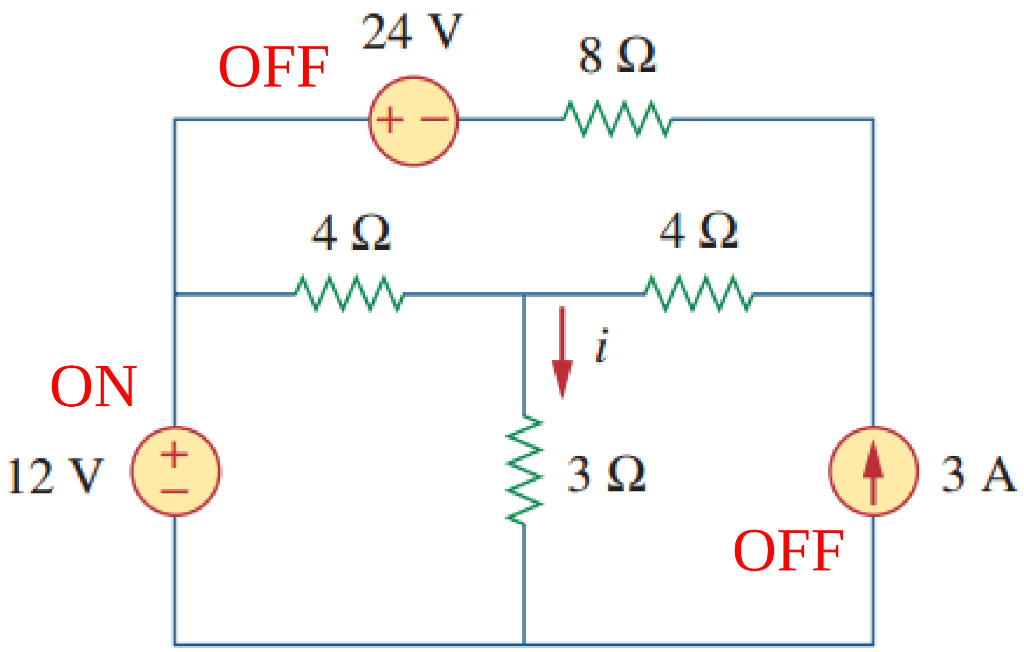
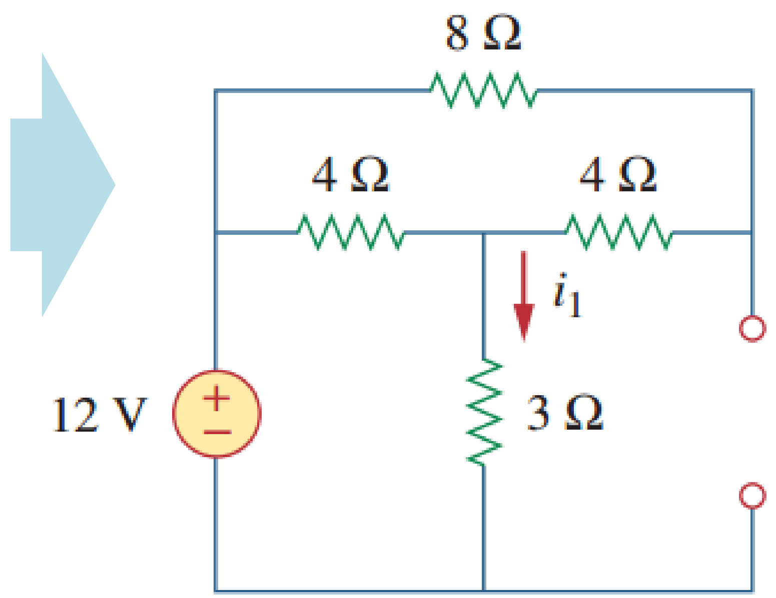
- Example 4.03

Example 4.03: Use the superposition theorem to find the *i* in the following circuits.



Solution: Step 1 turn off all sources but the 12 V one



$$i_1 = \frac{12}{4||(8+4)+3} = \frac{12}{3+3} = 2\text{ A}$$

4.3 Superposition Theorem (i)

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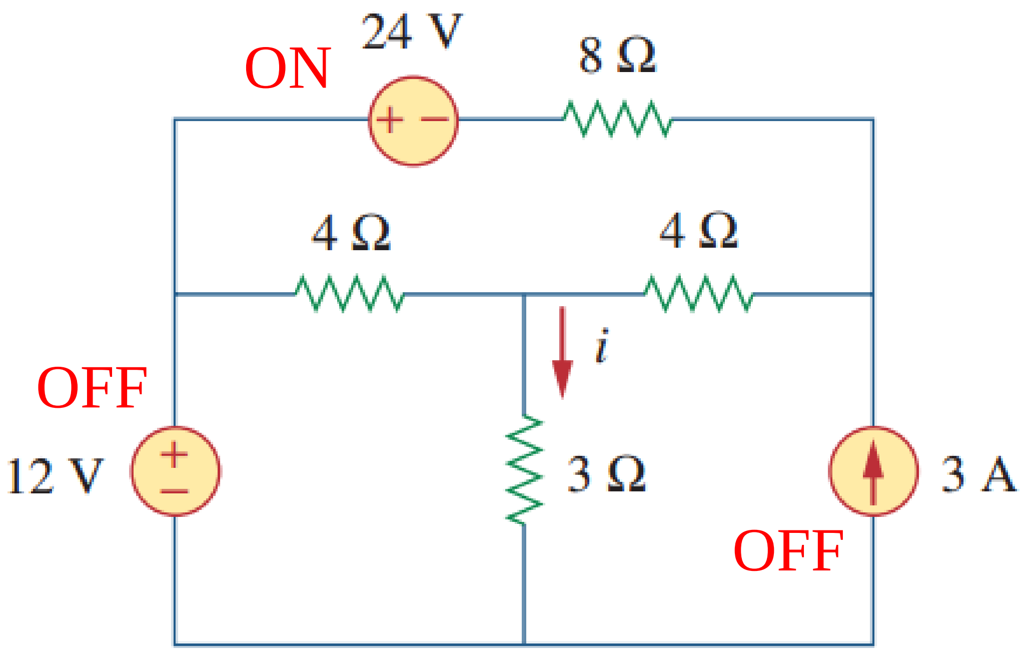
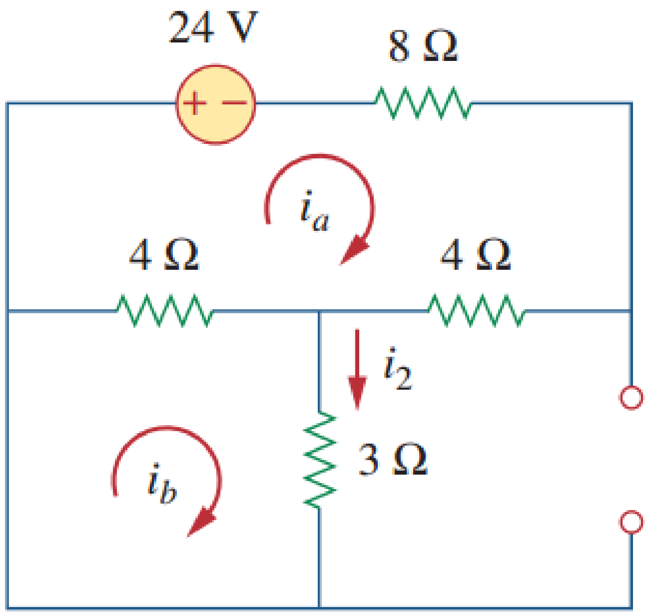
- Example 4.03

Example 4.03: Use the superposition theorem to find the i in the following circuits.



Solution: Step 2 turn off all sources but the 24 V one

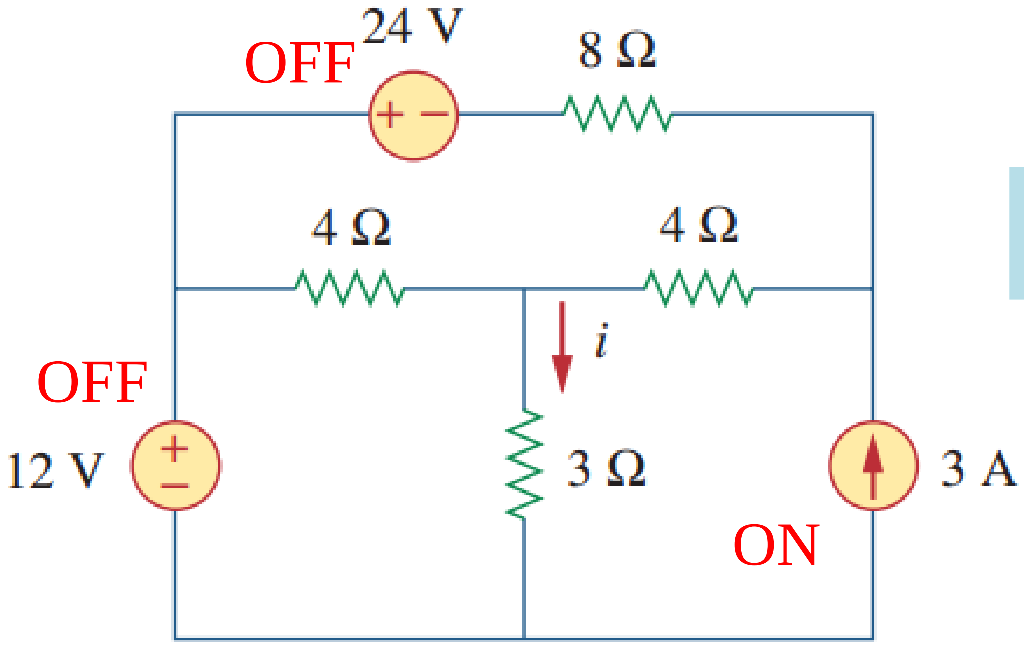


$$i_2 = -1 \text{ A}$$

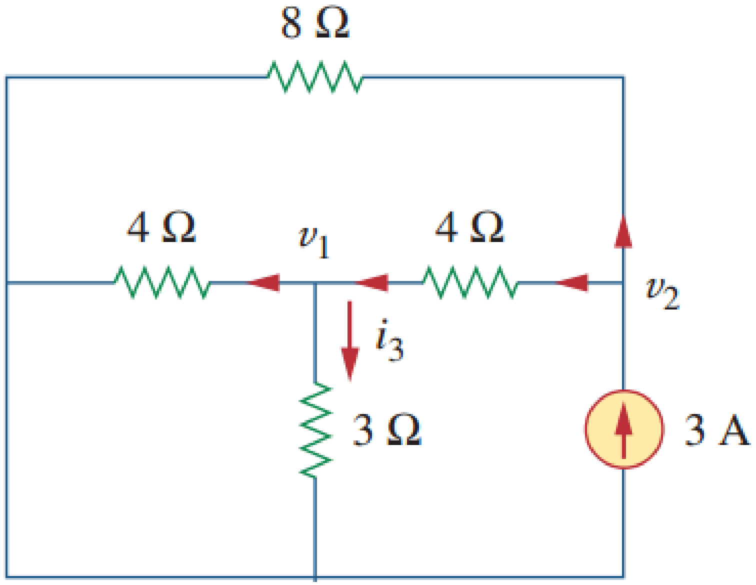
4.3 Superposition Theorem (j)

- Example 4.03

Example 4.03: Use the superposition theorem to find the i in the following circuits.



Solution: Step 3 turn off all sources but the 3 A one



$$i_3 = 1\ A$$

$$i = i_1 + i_2 + i_3$$

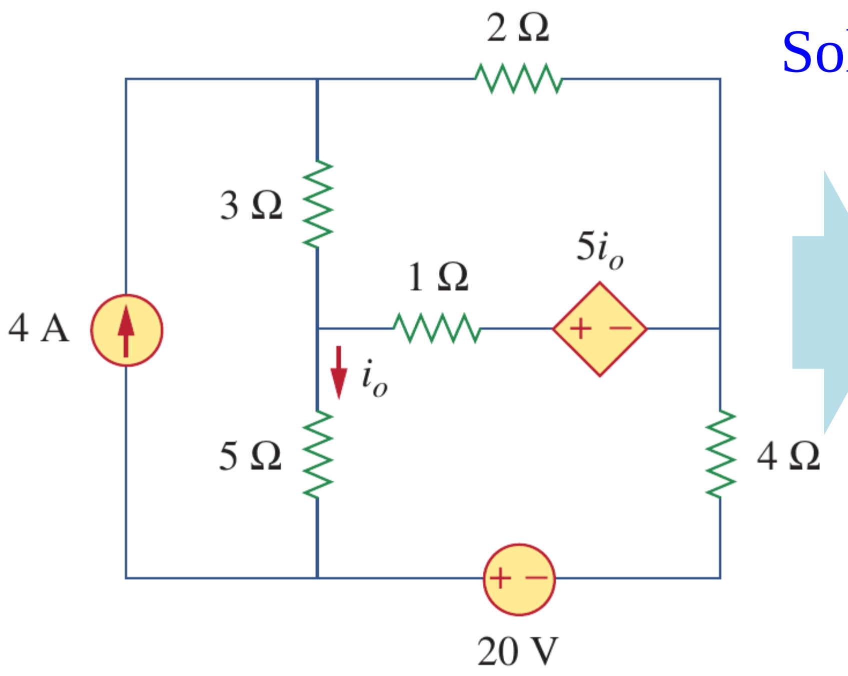
$$= 2 - 1 + 1$$

$$= 2\ \text{A}$$

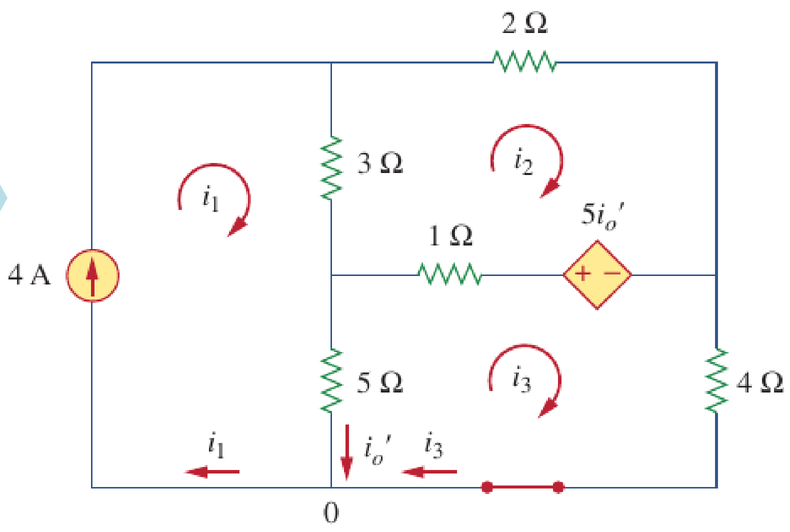
4.3 Superposition Theorem (k)

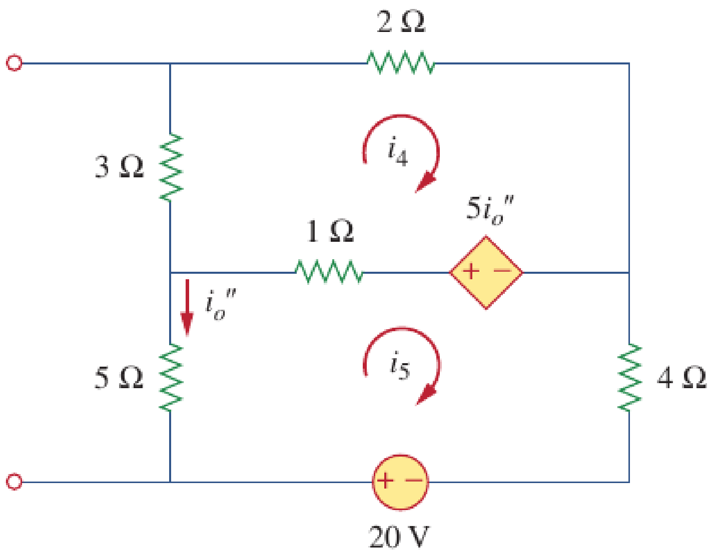
- Example 4.04 — superposition with dependent sources

Example 4.04: Use the superposition theorem to find the i_o in the following circuits.



Solution:





4.3 Superposition Theorem (I)

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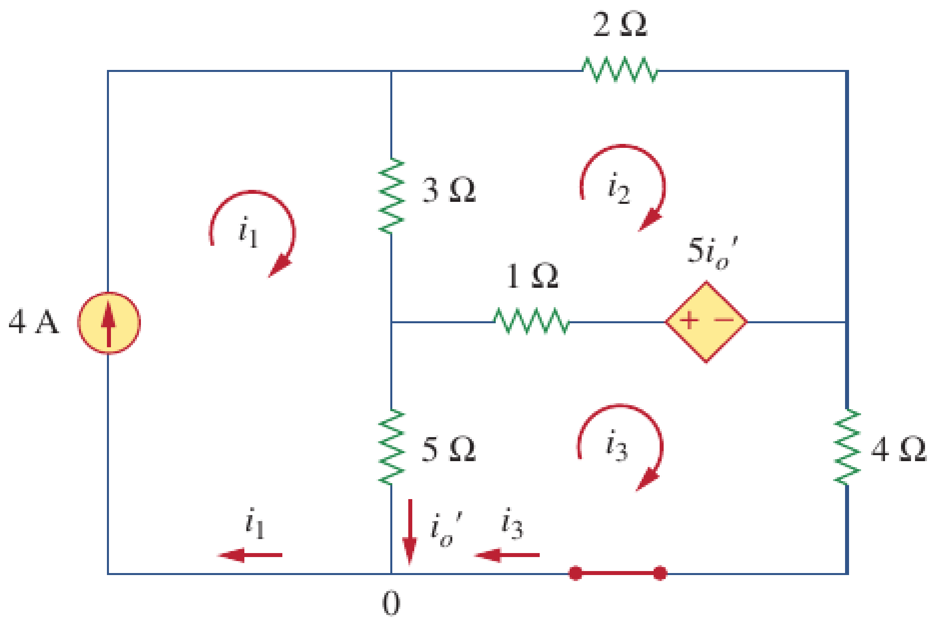
- Example 4.04 — superposition with dependent sources

Example 4.04: Use the superposition theorem to find the i_o in the following circuits.

Solution:



$For\ mesh$ 1 KVL: $i_1 = 4$ A

$For\ mesh$ 2 KVL: $6i_2 - 3i_1 - 1i_3 - 5i_o' = 0$

$For\ mesh$ 3 KVL: $10i_3 - 5i_1 - 1i_2 + 5i_o' = 0$

One node: $i_3 + i_o' - i_1 = 0$

So $i_o' = \dfrac{52}{17}$ A

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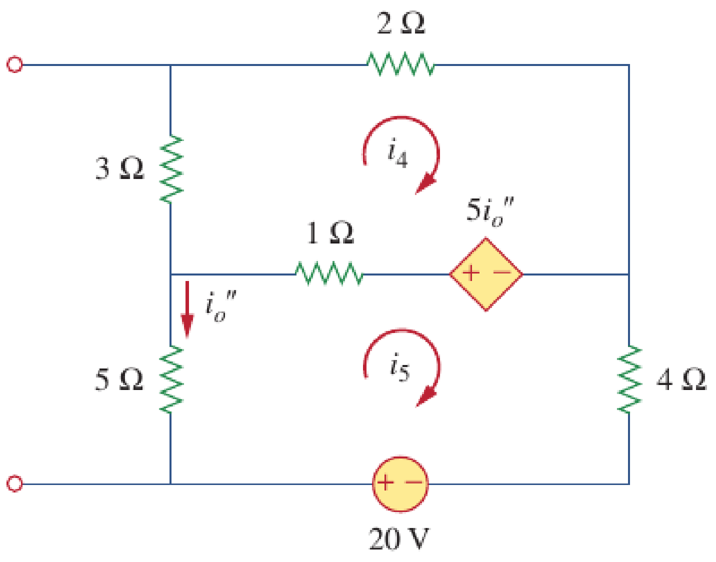
- Example 4.04 — superposition with dependent sources

Example 4.04: Use the superposition theorem to find the i_o in the following circuits.

Solution:



$$For\ mesh\ 4\ \text{KVL:}\quad 6i_4 - i_5 - 5i_o'' = 0$$

$$For\ mesh\ 5\ \text{KVL:}\quad 10i_5 - 1i_4 + 5i_o'' - 20 = 0$$

$$i_5 = -i_o''$$

$$\text{So } i_o'' = -\frac{60}{17}\ \text{A}$$

$$i_o = i_o' + i_o'' = \frac{52}{17} - \frac{60}{17} = -0.47\ \text{A}$$

4.3 Superposition Theorem (n)

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- Superposition and power

a) **POWER does not obey superposition!**

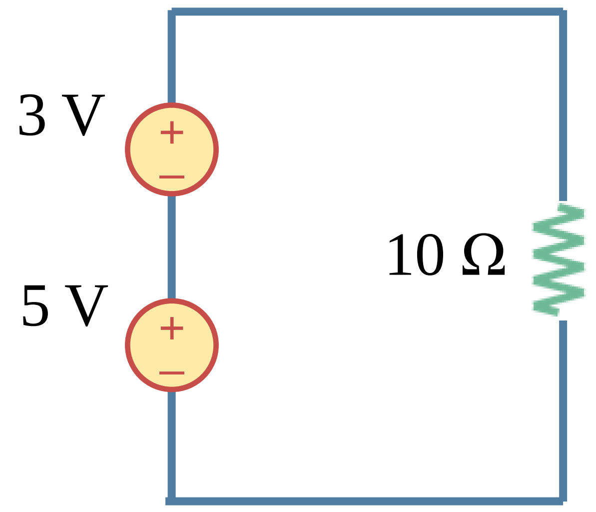
b) In the circuits shown in the right:

The power absorbed: $P = (3+5)^2/10 = 6.4$ W

If only the 3 V source: $P_1 = (3)^2/10 = 0.9$ W

If only the 5 V source: $P_2 = (5)^2/10 = 2.5$ W

c) $0.9 + 2.5 \neq 6.4$ W, $P_1 + P_2 \neq P$



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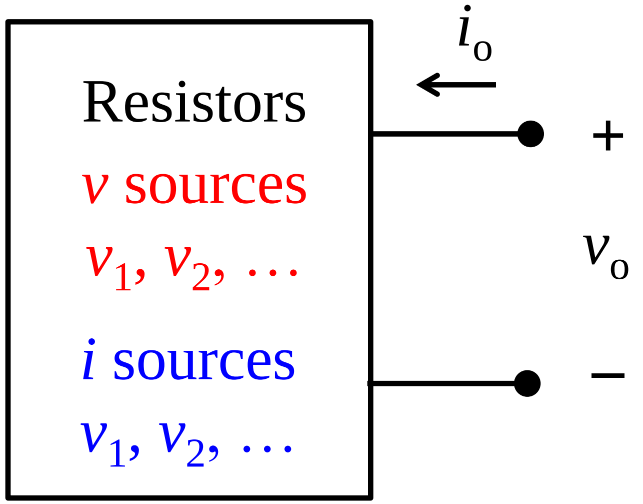
- Linearity theorem

1. For circuits containing linear elements and independent sources, any output voltage or current, denoted by variable y, is related linearly to the independent sources of the circuit. This is also true for circuit containing linear dependent sources.
2. $y = a_1v_1 + a_2v_2 + \ldots + a_mv_m + b_1i_1 + b_2i_2 + \ldots + b_ni_n$
3. where v_1 through v_m and i_1 through i_n are voltage and current values of the independent sources in the circuits, respectively, a_1 through a_m and b_1 through b_n are suitably dimensioned constants.



- $v_o = a_1v_1 + a_2v_2 + \ldots + a_mv_m$
  $+ b_1i_1 + b_2i_2 + \ldots + b_ni_n$
- $i_o = c_1v_1 + c_2v_2 + \ldots + c_mv_m$
  $+ d_1i_1 + d_2i_2 + \ldots + d_ni_n$

4.4 Homogeneity & Superposition (b)

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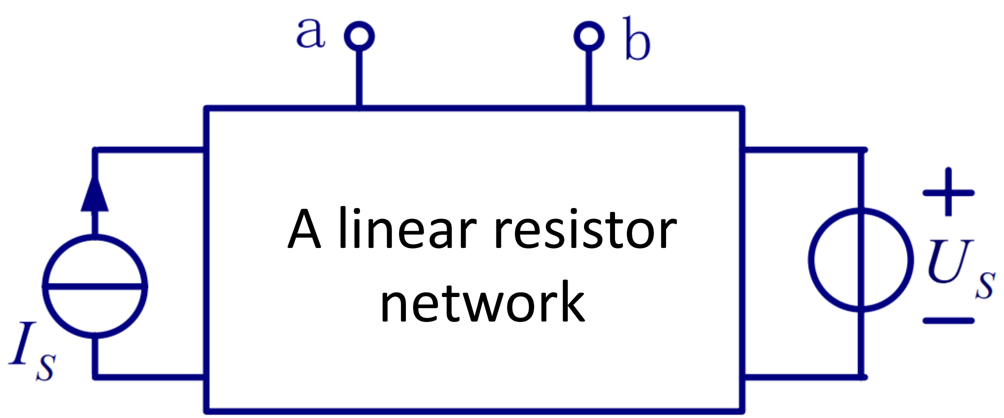
- Example 4.05

Example 4.04: A linear resistor network is as shown in the right figure, we know that:
V_{ab} = 0 V when V_s = 12 V and I_s = 2 A;
V_{ab} = 18 V when V_s = 24 V and I_s = 10 A;
Please find V_{ab} when V_s = 10 V and I_s = 3 A.



Solution: Remember that in a linear circuit: $y = a_1 v_1 + a_2 v_2 + \ldots + a_m v_m + b_1 i_1 + b_2 i_2 + \ldots + b_n i_n$

$$\begin{cases} 12a + 2b = 0 \\ 24a + 10b = 18 \end{cases} \Rightarrow \begin{cases} a = -0.5 \\ b = -3 \end{cases}$$

$$\Rightarrow \quad V_{ab} = -0.5V_s + 3I_s = 4\ V$$

Lesson 4 Circuit Theorems

4.1-4.4	Linear circuits and superposition theorem	Slides 08-31
4.5-4.6	**Source transformation theorem**	**Slides 33-47**
4.7	Thévenin’s theorem	Slides 49-60
4.8-4.9	Norton’s theorem	Slides 62-70
4.10	Maximum power transfer theorem	Slides 72-80
4.11	Summary and assignments	Slides 82-83

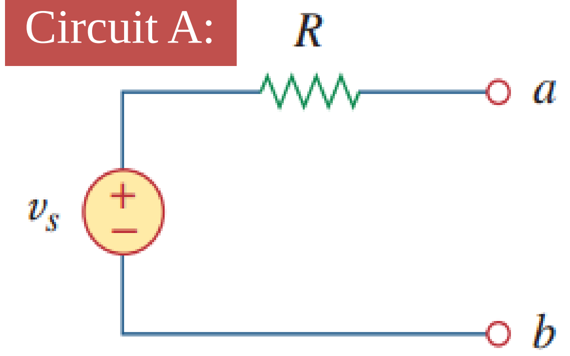
4.5 Source transformation (a)

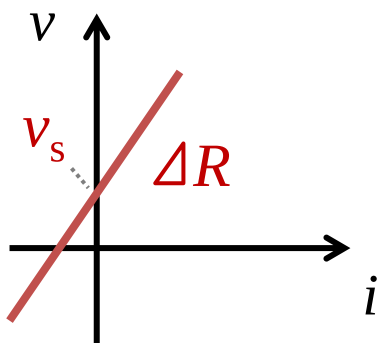
- Source transformation

a) Source transformation (电源变换) refers to the process of replacing a voltage source in series with a resistor by an equivalent circuits including a current source and a resistor, or vice versa.

b) Reminder: two circuits are equivalent when they share identical *i*-*v* characteristics (伏安特性)

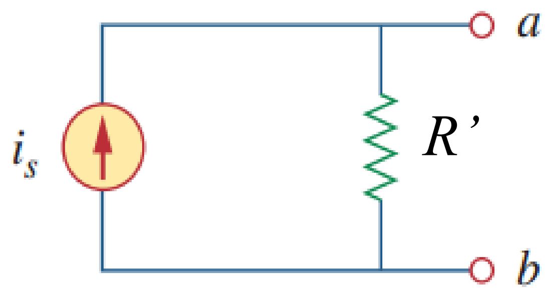
Circuit A:

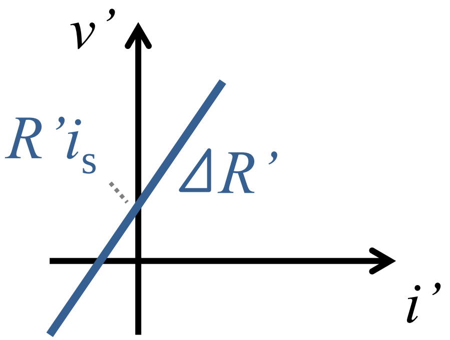




To make them equivalent: (1) $\boldsymbol{R = R'}$ & (2) $\boldsymbol{v_s = R \cdot i_s}$

Circuit B:





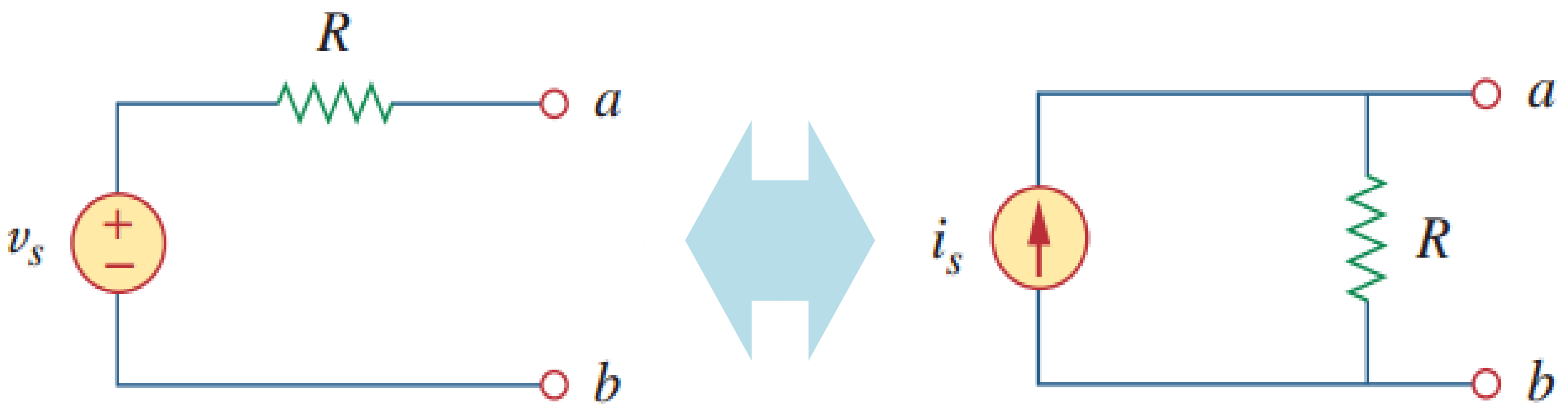
4.5 Source transformation (b)

- Important

a) $R = R'$ & $v_s = R \cdot i_s$

b) The direction of i_s points to the positive terminal of V-source.

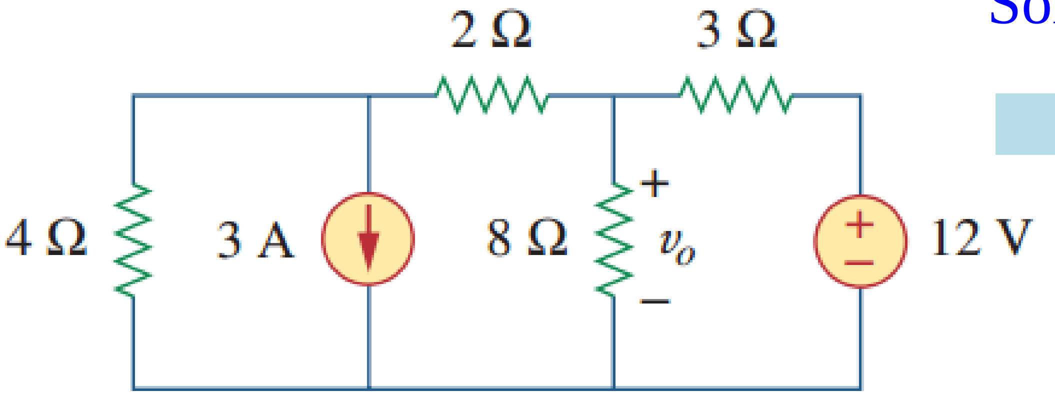
c) Source transformation also applies to dependent sources.



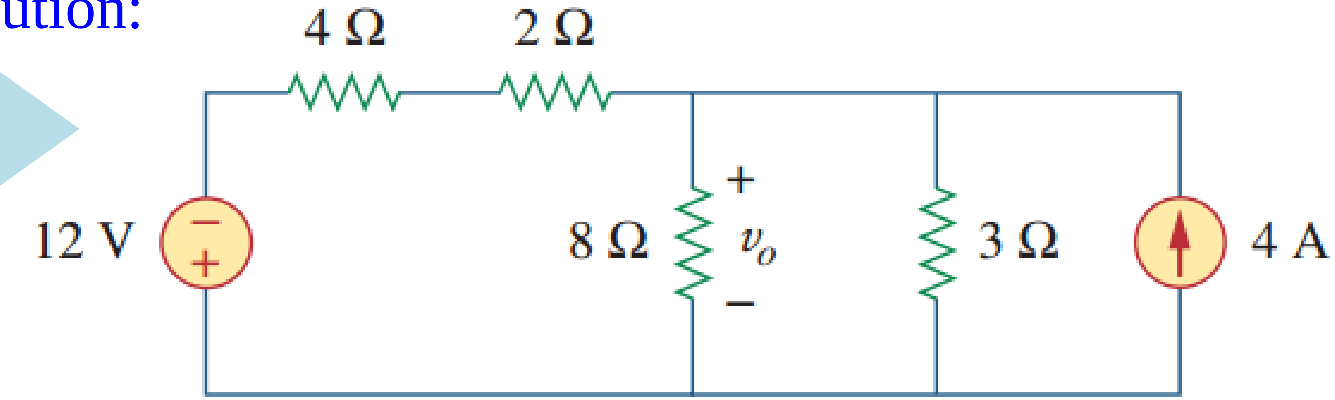
4.5 Source transformation (c)

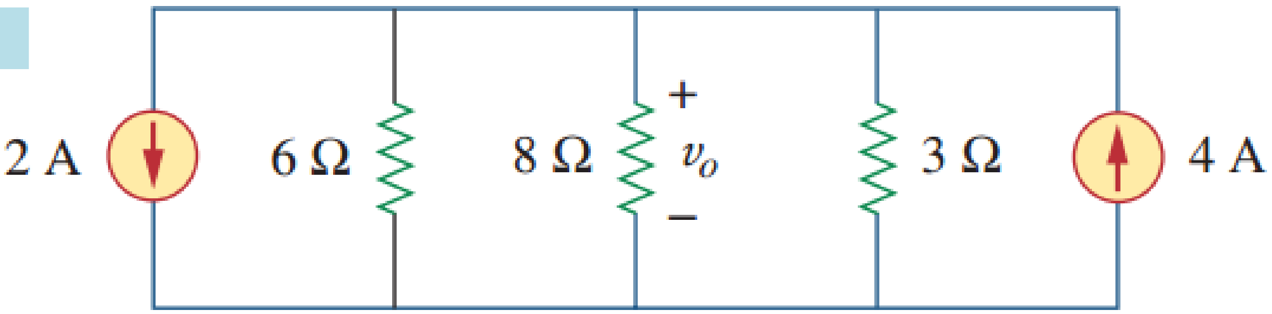
- Example 4.06

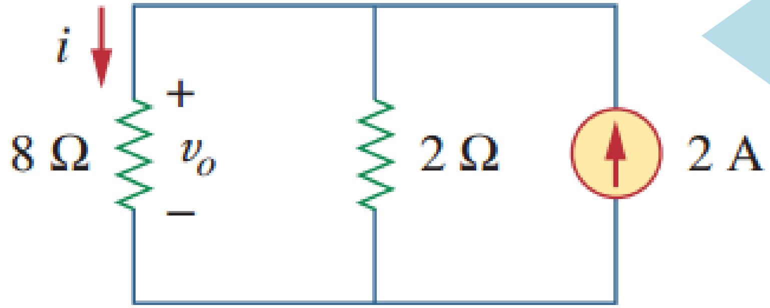
Example 4.06: Use the source transformation to find the v_o in the following circuits.



Solution:





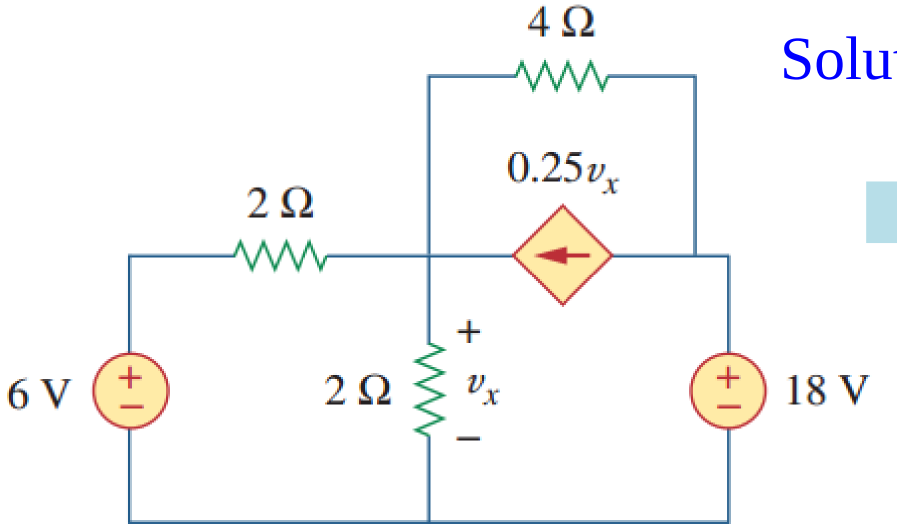


$v_0 = 3.2$ V

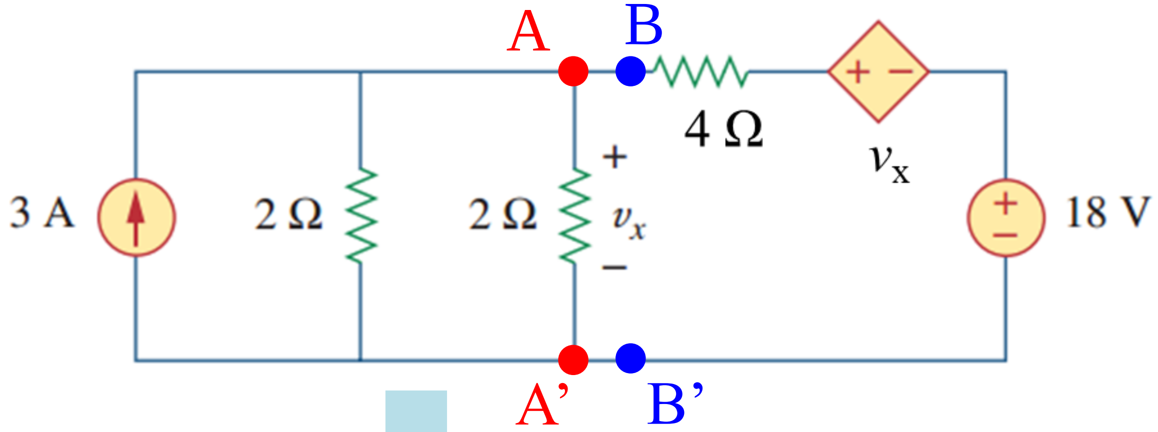
4.5 Source transformation (d)

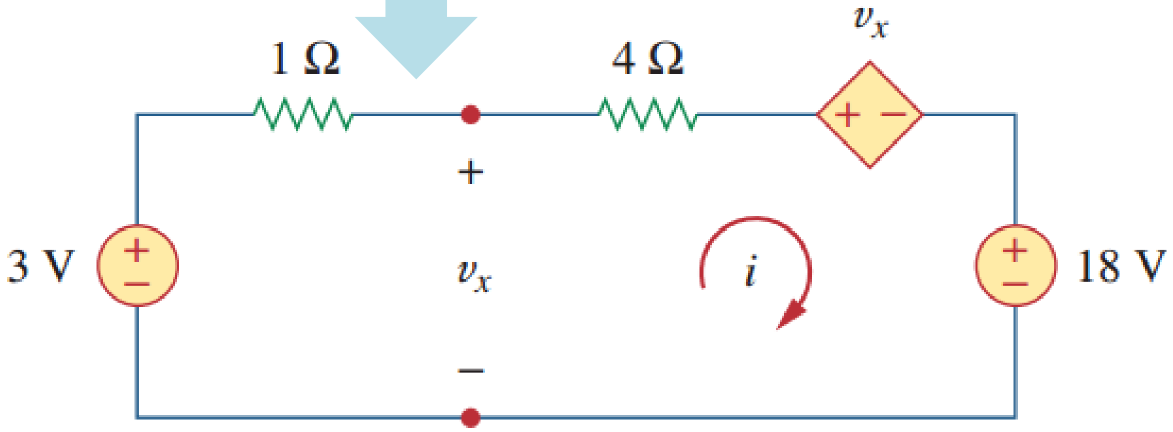
- Example 4.07

Example 4.07: Find v_x in the following circuits using source tranformation.



Solution:

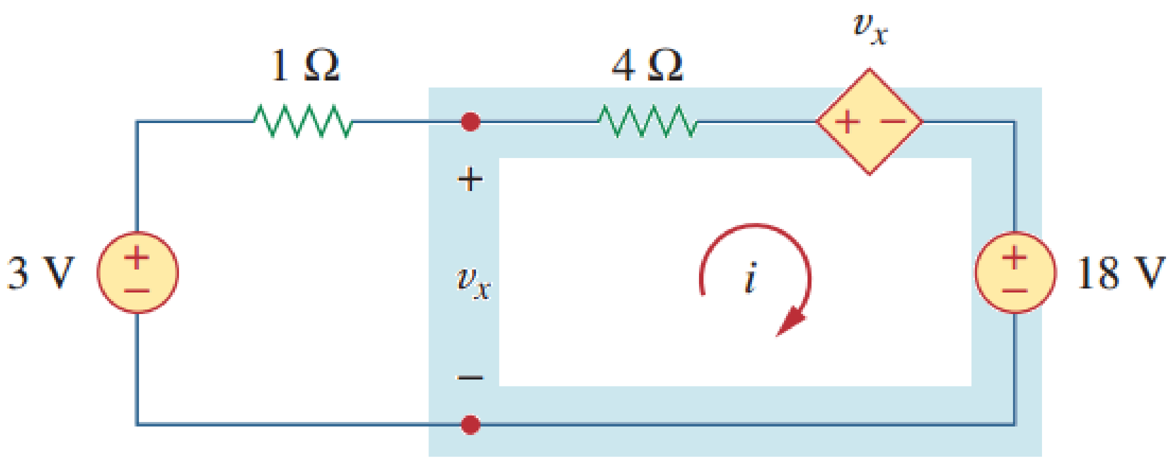




- Example 4.07

Example 4.07: Find v_x in the following circuits using source transformation.

Solution:



1. KVL for the entire loop:

$$-3 + (1+4)i + v_x + 18 = 0$$

2. KVL for the imaginary mesh:

$$-v_x + 4i + v_x + 18 = 0$$

3. Solve the equations:

$$i = -4.5\text{ A}, \quad v_x = 7.5\text{ V}$$

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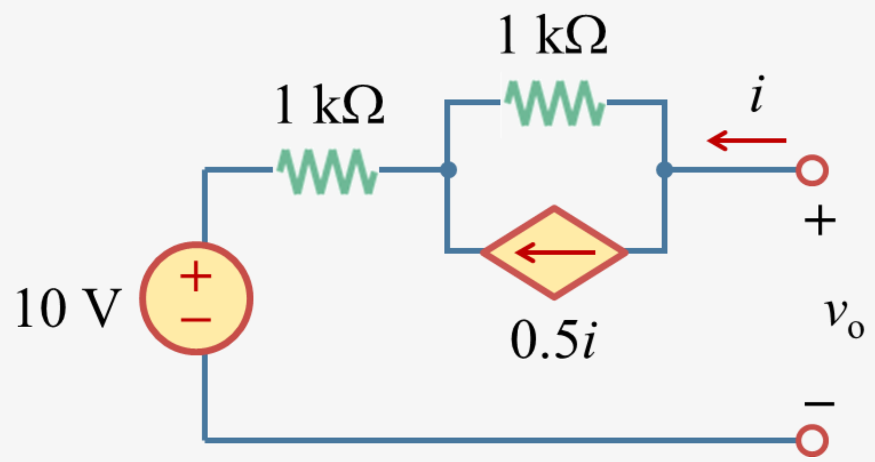
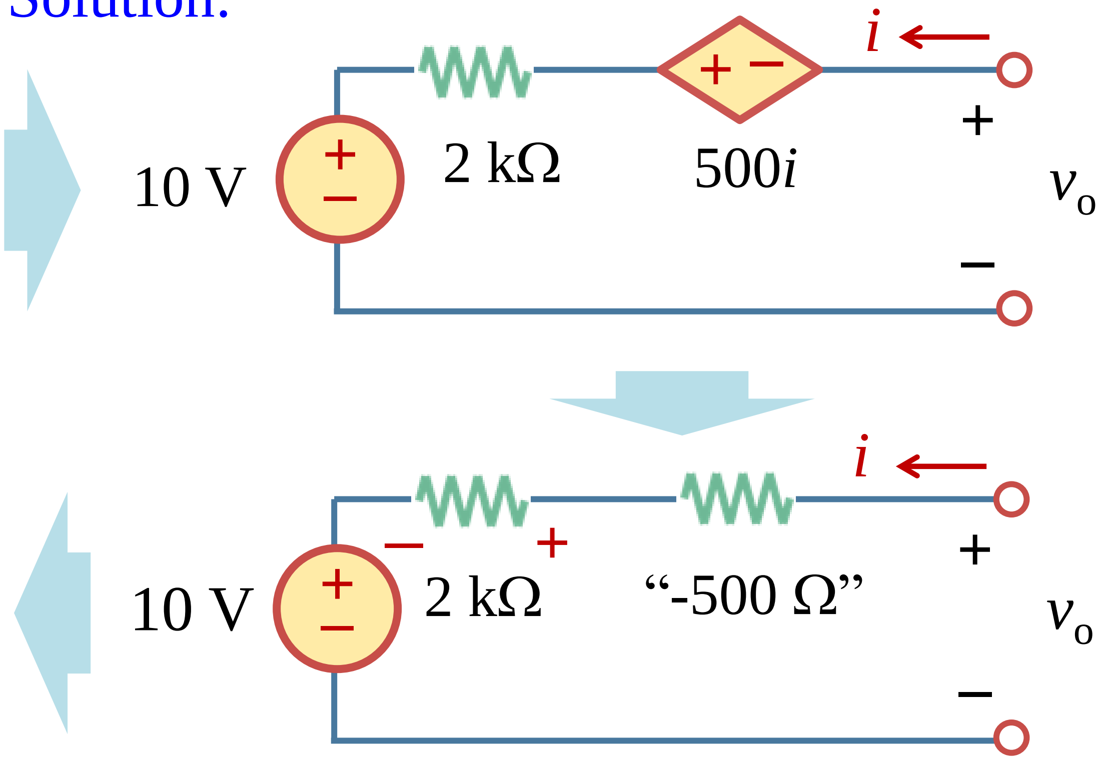
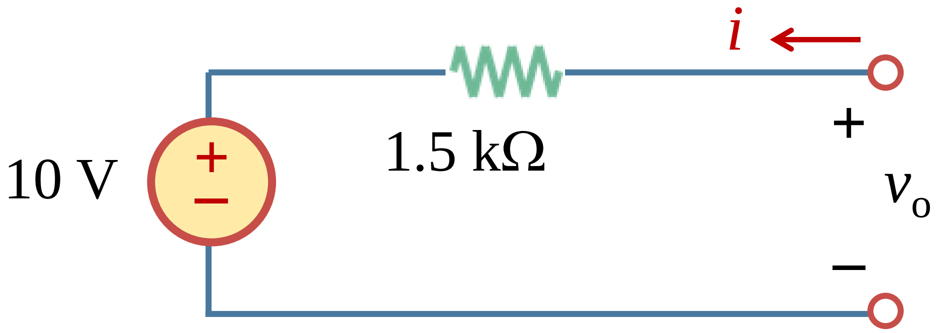
- Example 4.08

Example 4.08: Find the equivalent circuit which contains a voltage source in series with a resistor.



Solution:

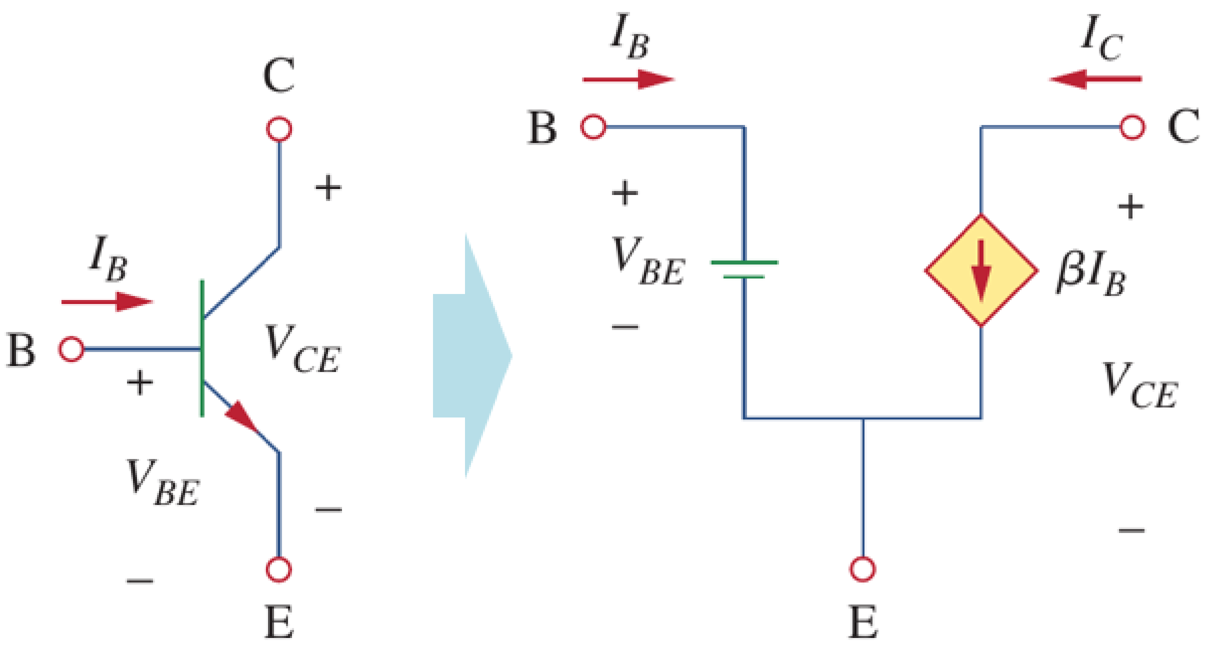




Supplementary: substitution theorem (替代定理)

a) Substitution Theorem states that the voltage across any branch or the current through that branch of a network being known, the branch can be replaced by the combination of various elements that will make the same voltage and current through that branch. It works for both linear and nonlinear circuits.

b) Recall: the DC equivalent circuit of a BJT (Lesson 3).



4.6 More examples (a)

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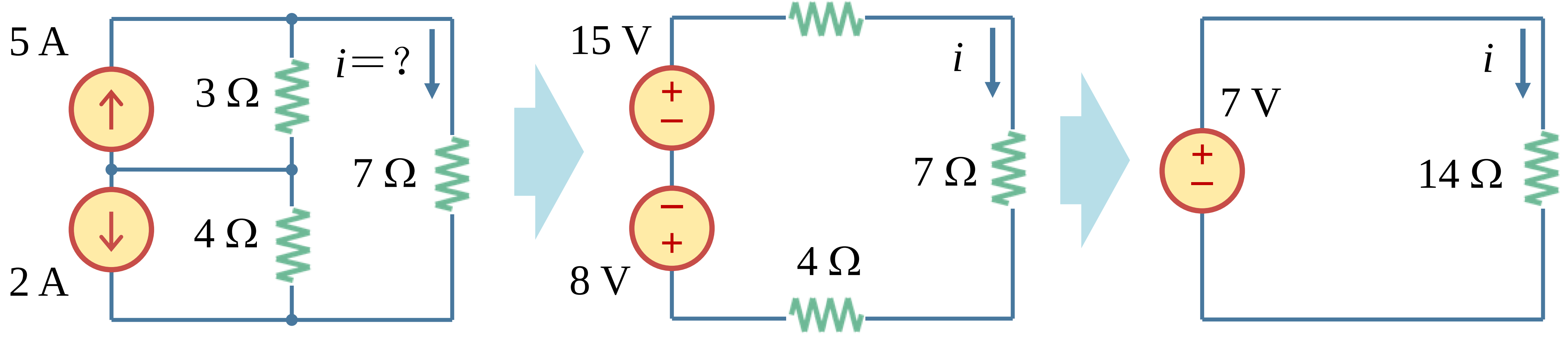
- Example 4.09

Example 4.09: Find i in the following circuit using source transformation.

Solution:



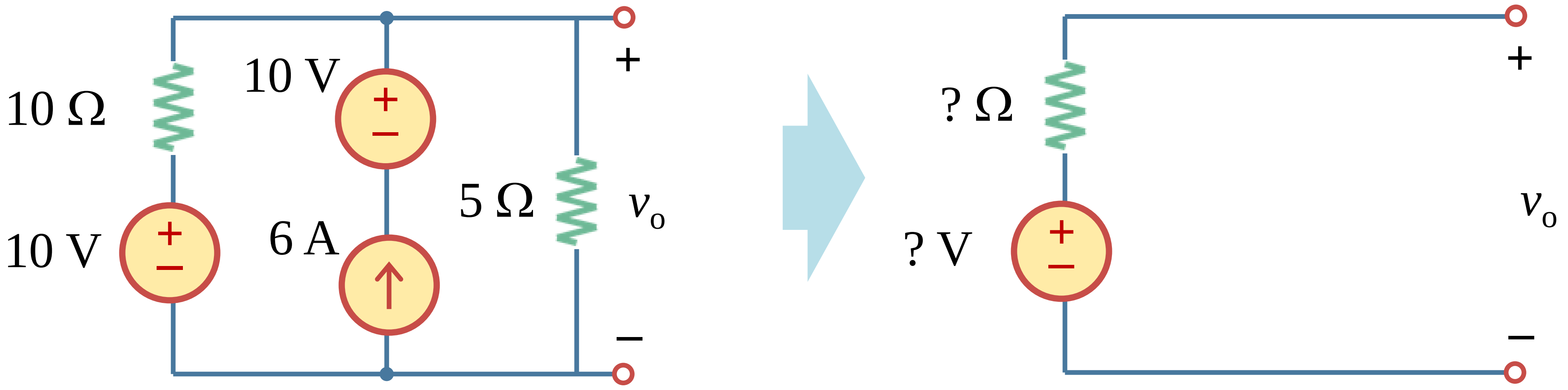
$i = 7 / 14 = 0.5$ A

4.6 More examples (b)

- Example 4.10

Example 4.10: Find the equivalent circuit which contains a voltage source in series with a resistor.

Solution:

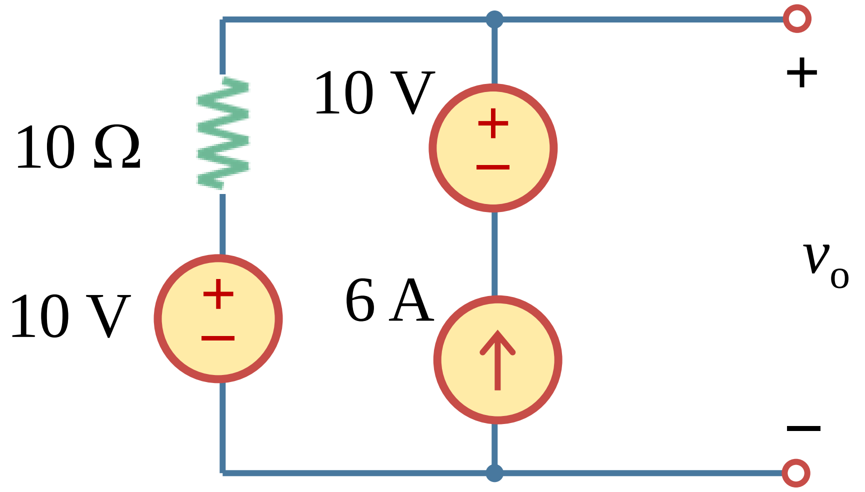


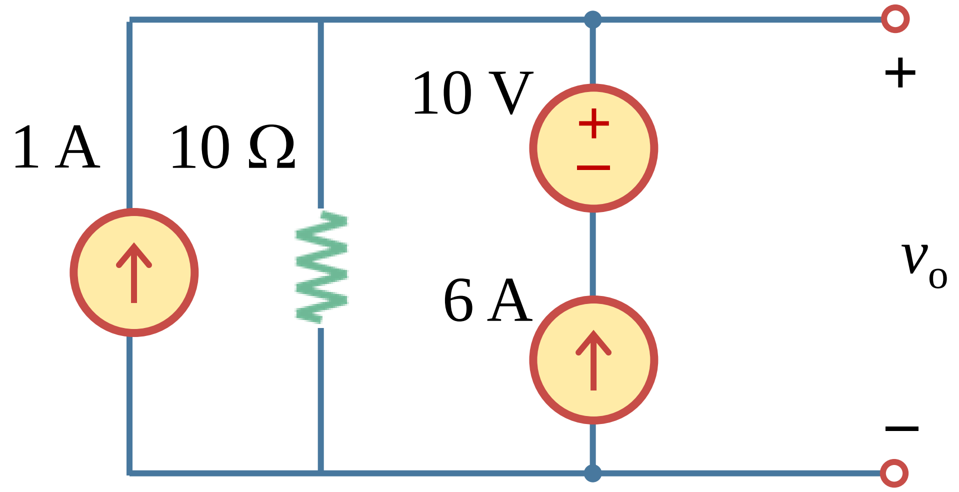
4.6 More examples (c)

- Example 4.10

Example 4.10: Find the equivalent circuit which contains a voltage source in series with a resistor.

Solution: Step 1 change the left 10 V source + the resistor into an equivalent current source in parallel with a resistor.





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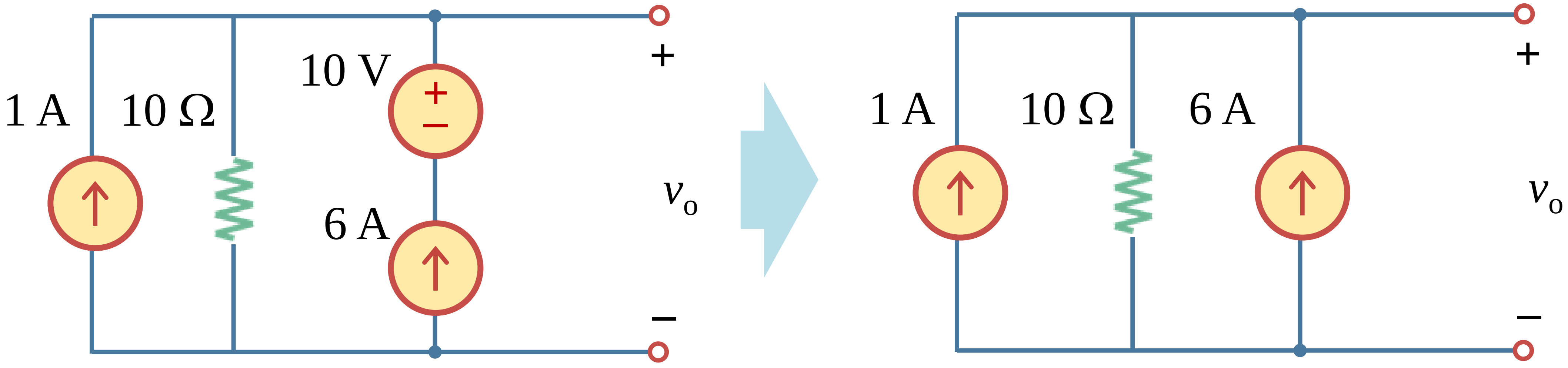
- Example 4.10

Example 4.10: Find the equivalent circuit which contains a voltage source in series with a resistor.

Solution: Step 2 change “voltage-and-current-sources-in-series” branch into an equivalent current source.



4.6 More examples (e)

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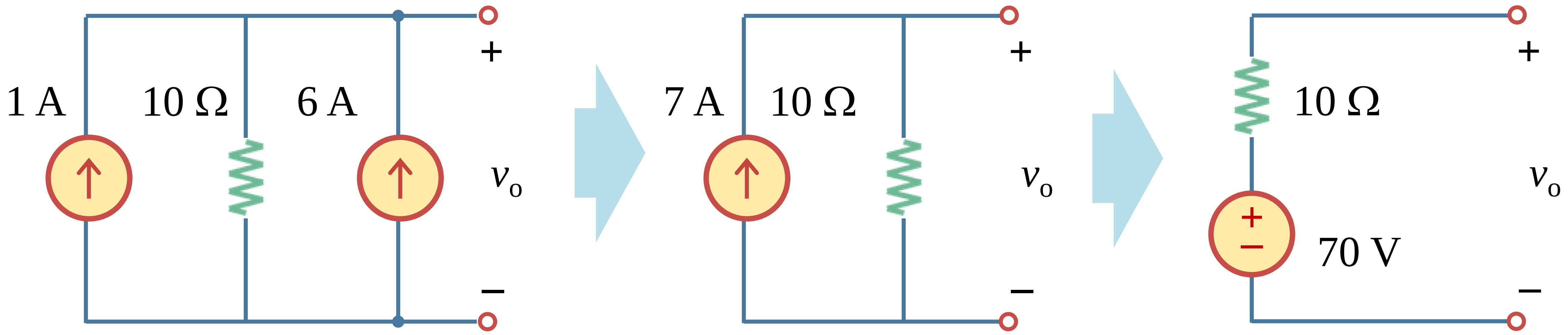
- Example 4.10

Example 4.10: Find the equivalent circuit which contains a voltage source in series with a resistor.

Solution: Step 3 combine the current sources and change them and the parallel resistor into a voltage source and a resistor in series .



4.6 More examples (f)

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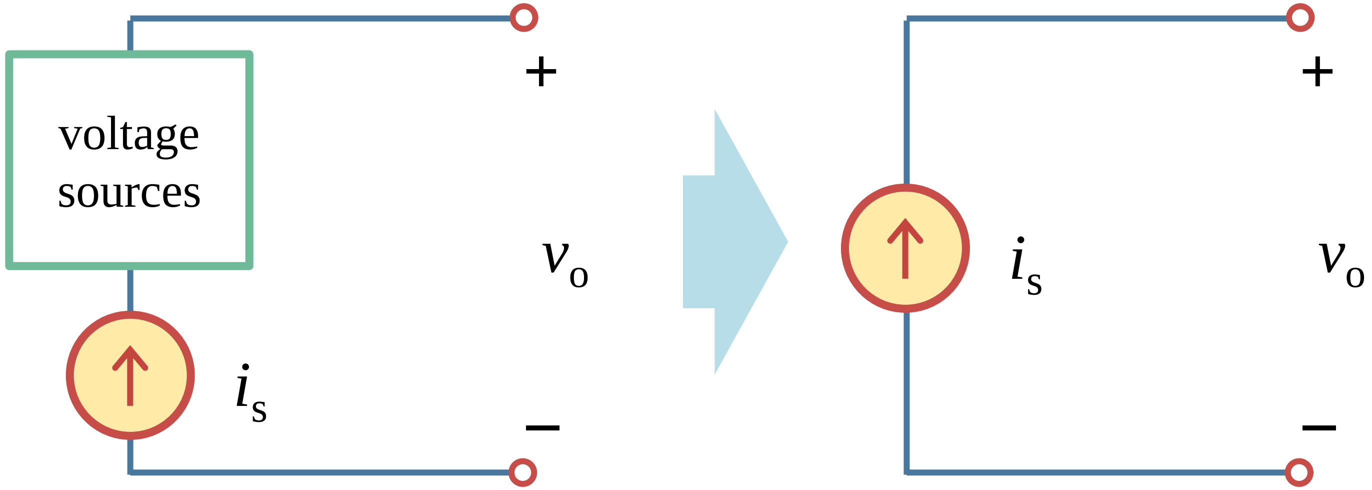
- Equivalent circuits of "a current source in series with an element"

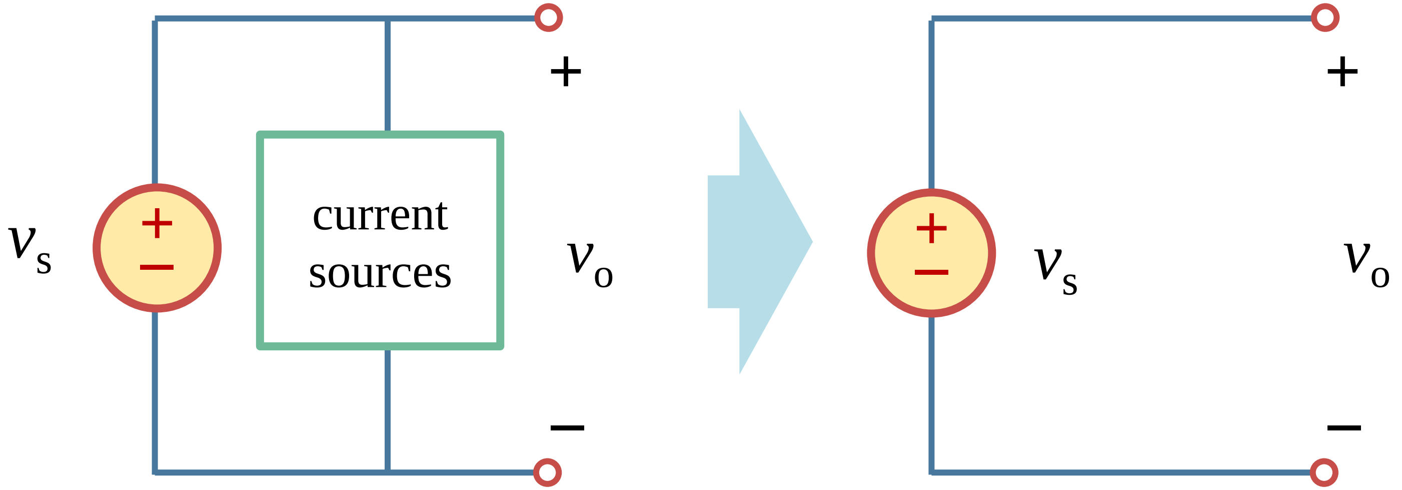


a) If a source unit includes a current source in series with a resistor or a voltage source, then one of the equivalent circuits of this unit is simply the current source. (Substitution theorem)

b) DO NOT connect two ideal current sources in series, unless they export the same current.

4.6 More examples (g)

- Equivalent circuits of "a voltage source in parallel with an element"



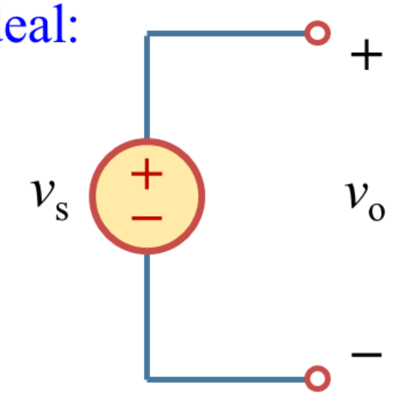
a) If a source unit includes a votlage source in parallel with a resistor or a current source, then one of the equivalent circuits of this unit is simply the voltage source. (Substitution theorem)

b) DO NOT connect two ideal voltage sources in parallel, unless they export the same current.

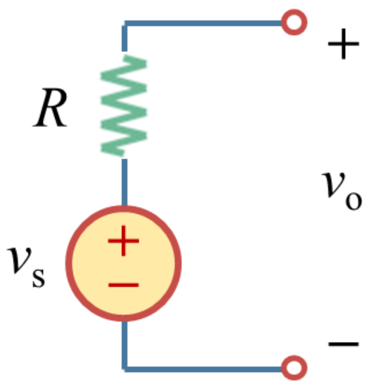
4.6 More examples (h)

- Ideal and Practical sources

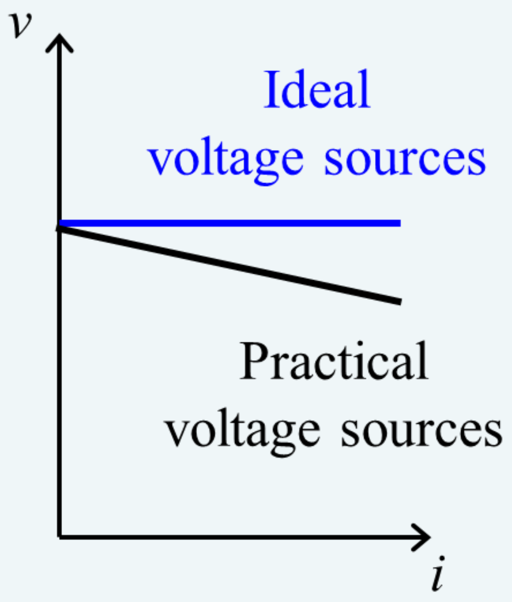
Ideal:



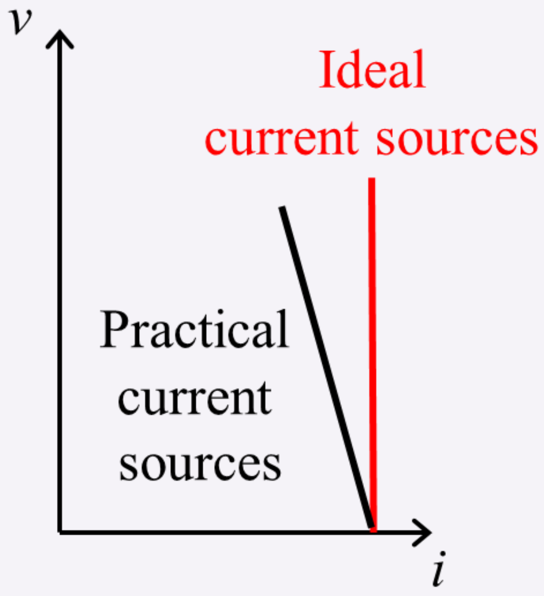
Practical:



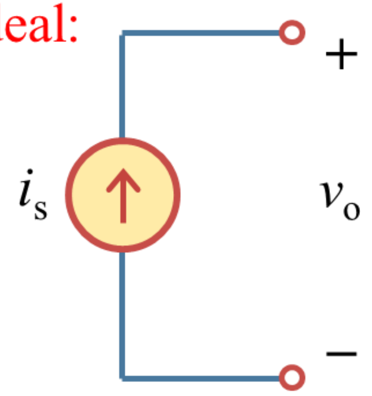
Votlage sources



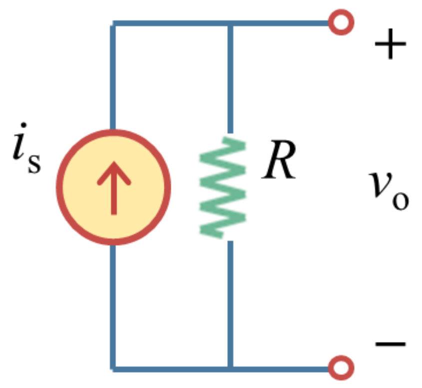
Current sources



Ideal:



Practical:



Lesson 4 Circuit Theorems

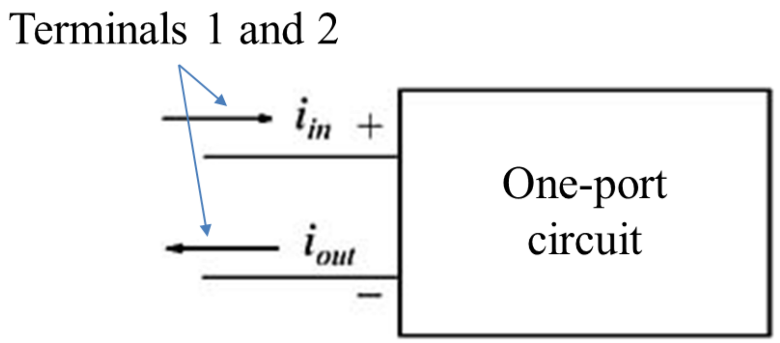
4.1-4.4	Linear circuits and superposition theorem	Slides 08-31
4.5-4.6	Source transformation theorem	Slides 33-47
4.7	**Thévenin’s theorem**	**Slides 49-60**
4.8-4.9	Norton’s theorem	Slides 62-70
4.10	Maximum power transfer theorem	Slides 72-80
4.11	Summary and assignments	Slides 82-83

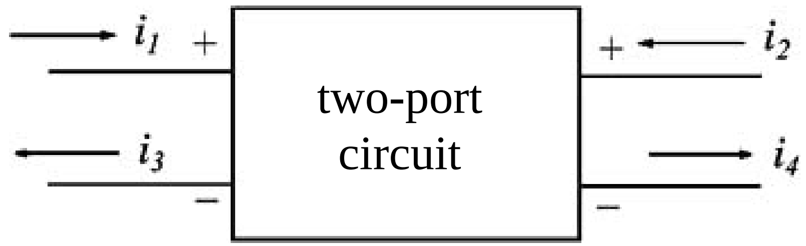
4.7 Thévenin's theorem (a)

- Recall: two-terminal networks

a) One-port network (two-terminal network) is generally a network that has two pairs of terminals. One-port circuits are usually found at the beginning and the end of a circuit. (Port 端口； terminal 端子)

b) A two-port circuit has 4 terminals (2 pairs). one pair is denoted as the input terminals, and the other as the output terminals.





4.7 Thévenin's theorem (b)

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- Background

a) Oftentimes you would need to know the *i* and *v* at a single branch while designing or analyzing a circuit. In this scenario, the other portion of the circuit can be considered as a two-terminal building block.

b) If the building block is based on linear elements like resistors, plus independent sources and linear dependent sources, then this build block can be considered as a network containing either "a voltage source in series with a resistor" or "a current source in parallel with a resistors."

c) Thévenin's theorem & Norton theorem

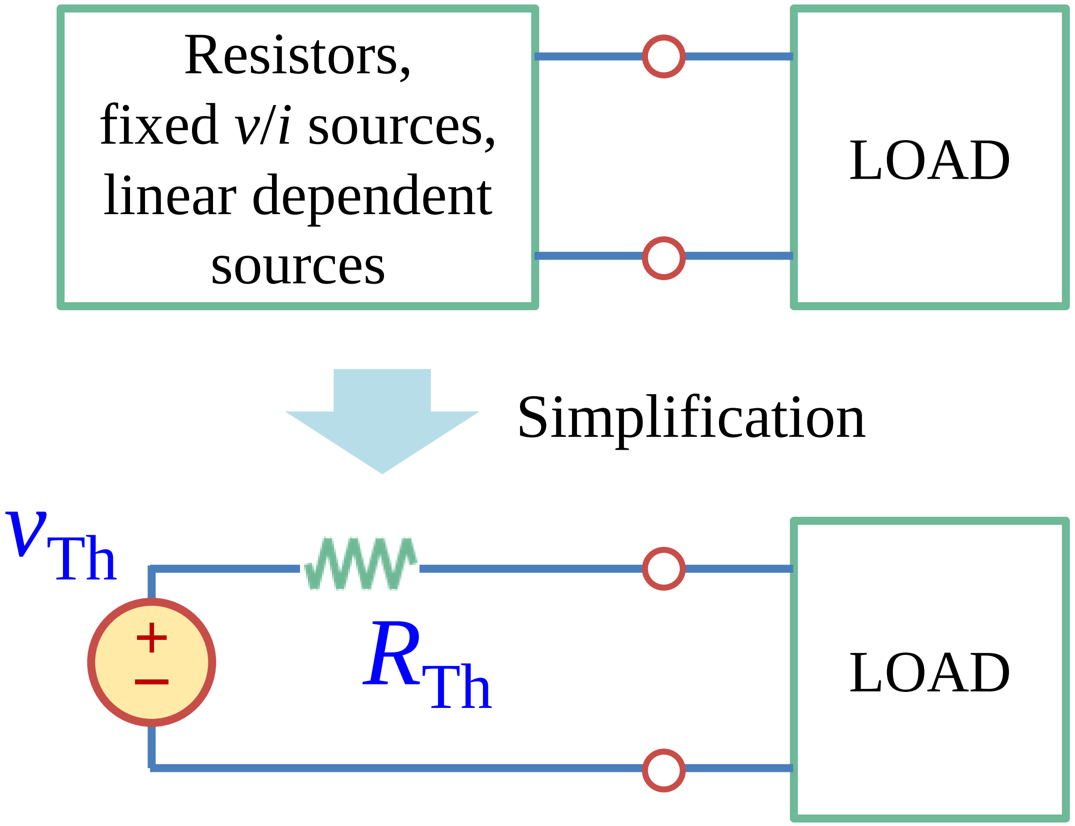
4.7 Thévenin's theorem (c)

- Thévenin's theorem (戴维南定理)

a) Thévenin's theorem: Any two-terminal network consisting of resistors, fixed voltage/current sources and linear dependent sources is externally equivalent to a circuit consisting of a resistor (R_{Th}) in series with a fixed voltage source (v_{Th}).

b) v_{Th} is the open-circuit voltage (without the load) at the terminals.

c) R_{Th} is the input or the equivalent resistance measured at the terminals when all the internal independent sources are turned off.



4.7 Thévenin's theorem (d)

- Thévenin circuit properties

a) A Thévenin equivalent circuit has a straight line characteristic with the equation:

$$v_o = R_{Th} \cdot i_o + V_{Th}$$

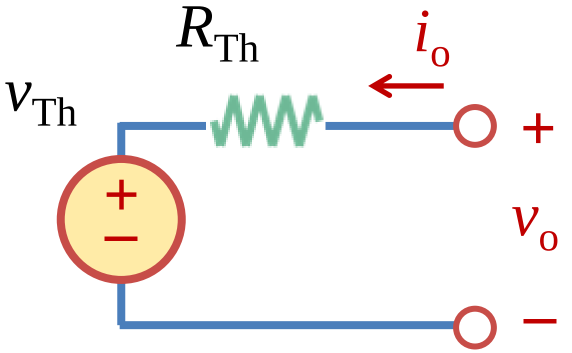
a) There are three important quantities:

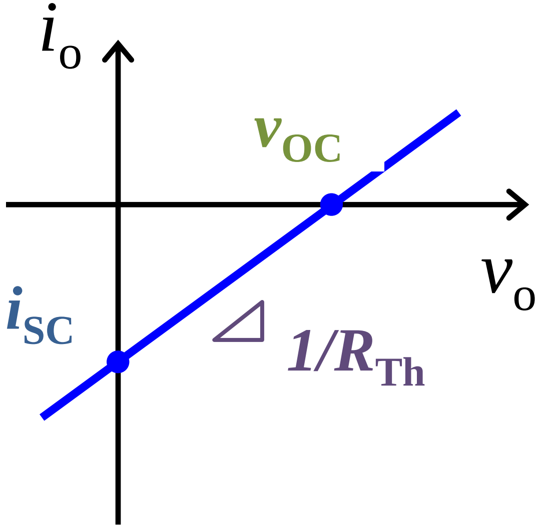
Open-circuit voltage v_{OC}: If $i_o = 0$ V, $v_o = v_{Th} = v_{OC}$

Short-circuit current i_{SC}: If $v_o = 0$ V, $i_o = -i_{sc} = -v_{Th}/R_{Th}$

Thévenin resistance v_{OC}: $R_{Th} = v_{OC}/i_{SC} = R_{eq}$

a) In any two-terminal circuit with the same characteristic, the three quantities will have the same values. So if we can determine two of them, we can work out the Thévenin equivalent.





4.7 Thévenin's theorem (e)

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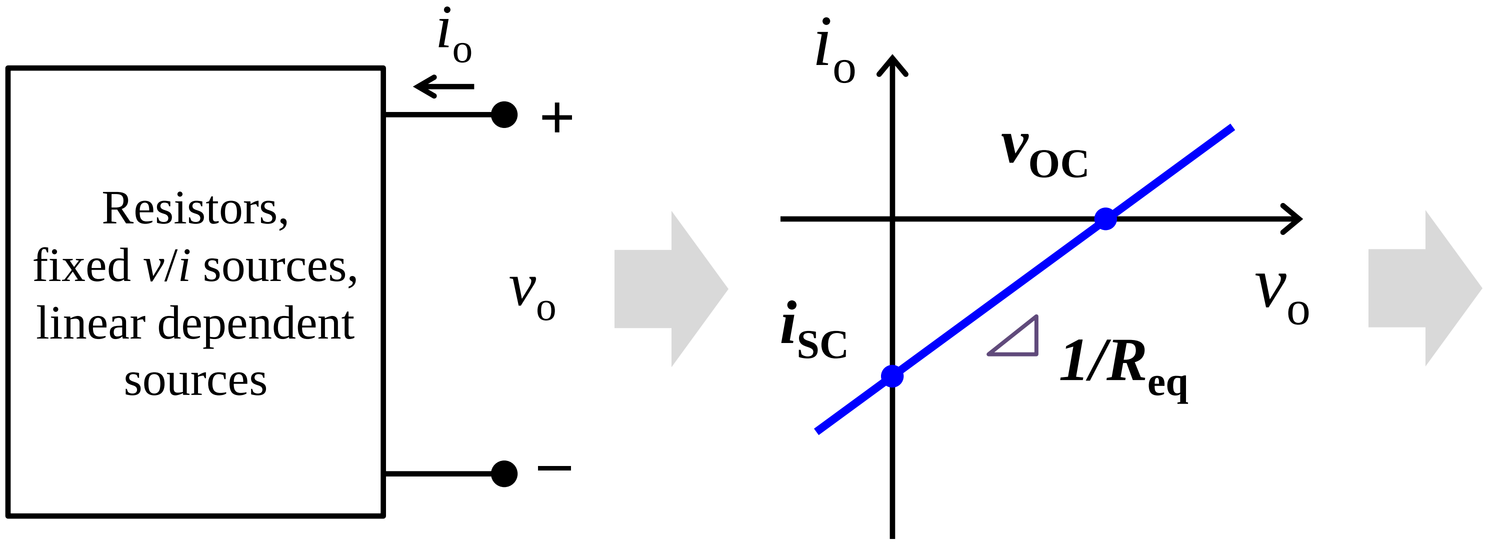
- Why v_{oc}, i_{sc} and R_{eq}? To determine the *i*-*v* characteristics at the port.



$$i_o = v_o/R_{eq} - i_{sc}$$

Or

$$v_o = R_{eq} \cdot i_o + V_{oc}$$

a) When you want to substitute a certain circuit by its equivalent circuit, in order to simplify a circuit, you need to first know need to know *i*-*v* characteristics of the original circuit.

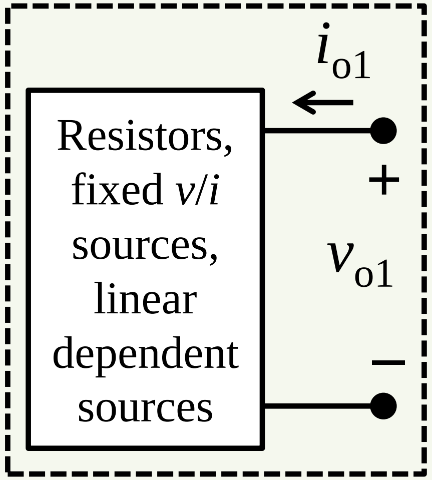
b) For a circuit based on linear elements, independent sources and linear dependent sources, v_{oc}, i_{sc} and R_{eq} are parameters to determine the *i*-*v* characteristics at the port.

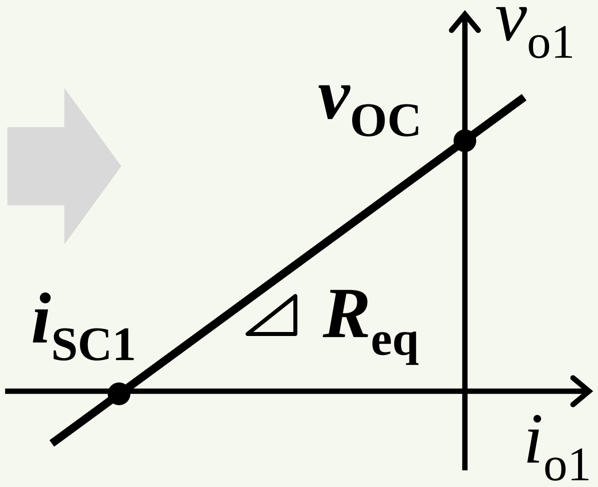
c) THIS IS WHY YOU NEED THEM IN THEVENIN AND NORTON THEOREMS !

4.7 Thévenin's theorem (f)

- Thévenin's theorem

The original circuit





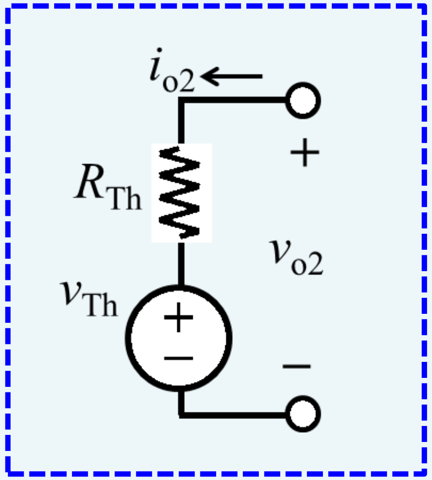
$$v_{o1} = i_{o1}R_{eq} + v_{oc}$$

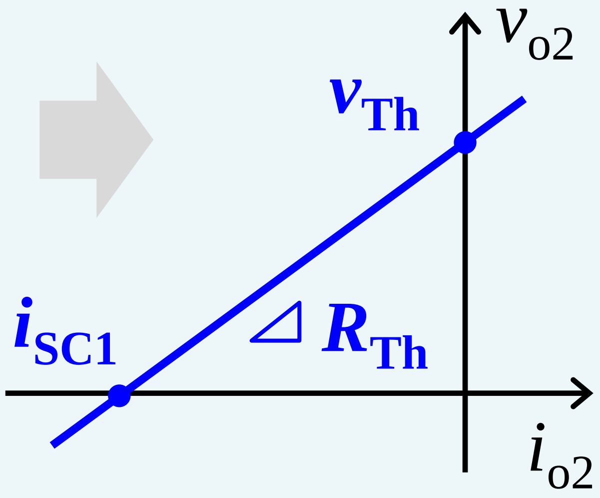
To substitute

We make them equivalent:

$$v_{Th} = v_{oc} \ \& \ R_{Th} = R_{eq}$$

The equivalent circuit



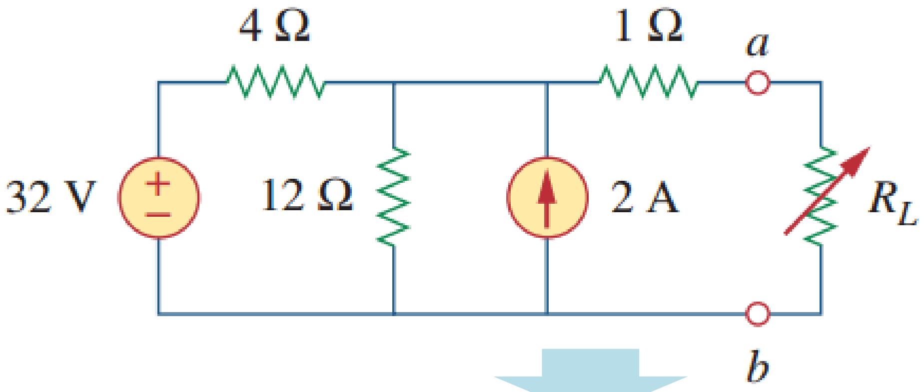


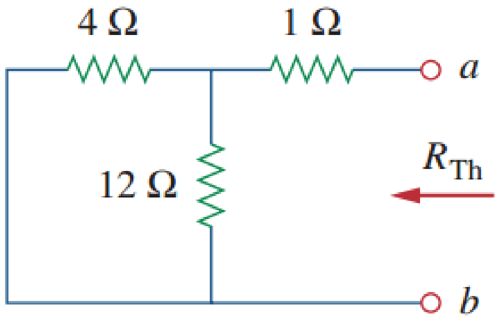
$$v_{o2} = i_{o2}R_{Th} + v_{Th}$$

4.7 Thévenin's theorem (g)

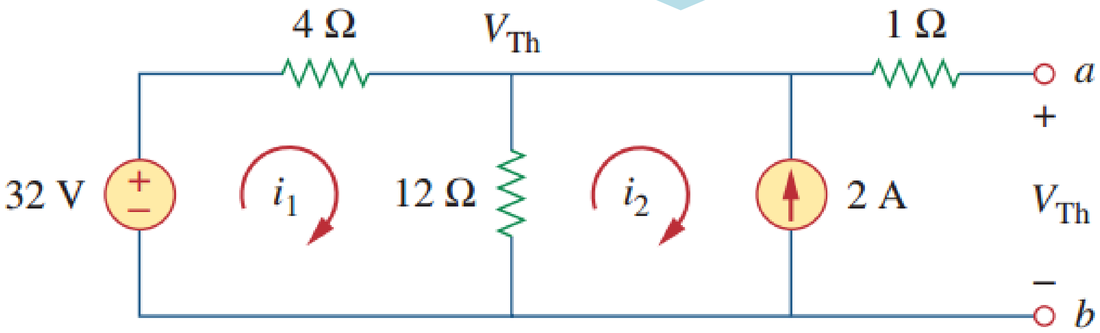
- Example 4.11

Example 11: Find the Thévenin equivalent circuit to the left of the terminals a-b. Then find the current through R_L=6, 16 and 36 Ω.

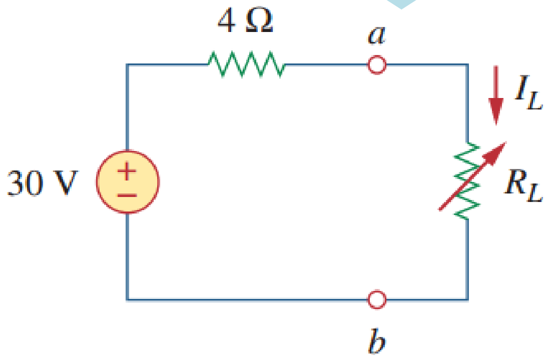




1. Turn off all independent sources and find R_{Th}



2. Find the open-circuit voltage to get V_{Th}



3. Thevenin's equivalent

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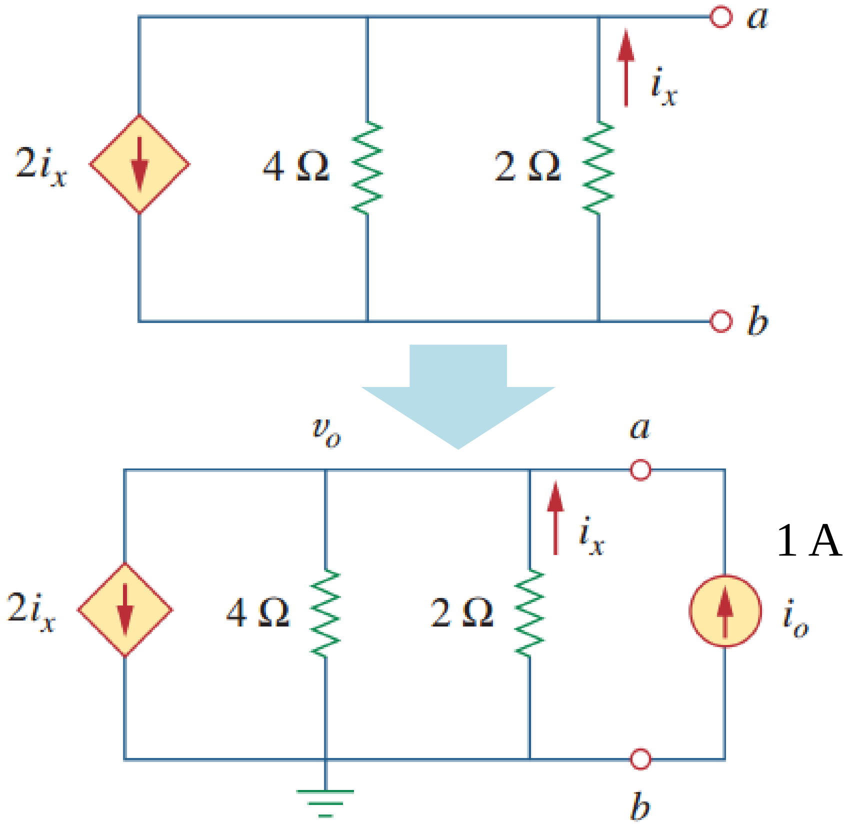
- Example 4.12 (When there are no independent sources)

Example 4.12: Determine the Thevenin equivalent of the circuit at terminals a-b.



Solution:

- $v_{Th} = 0$ V since there is no independent sources

- to find out R_{Th}

Assume a current source (1 A) connected to the port ab. Apply KCL at node a gives:

$1 - 2i_x + i_x - v_o/4 = 0$, while $v_o = v_{ab} = -2i_x$.

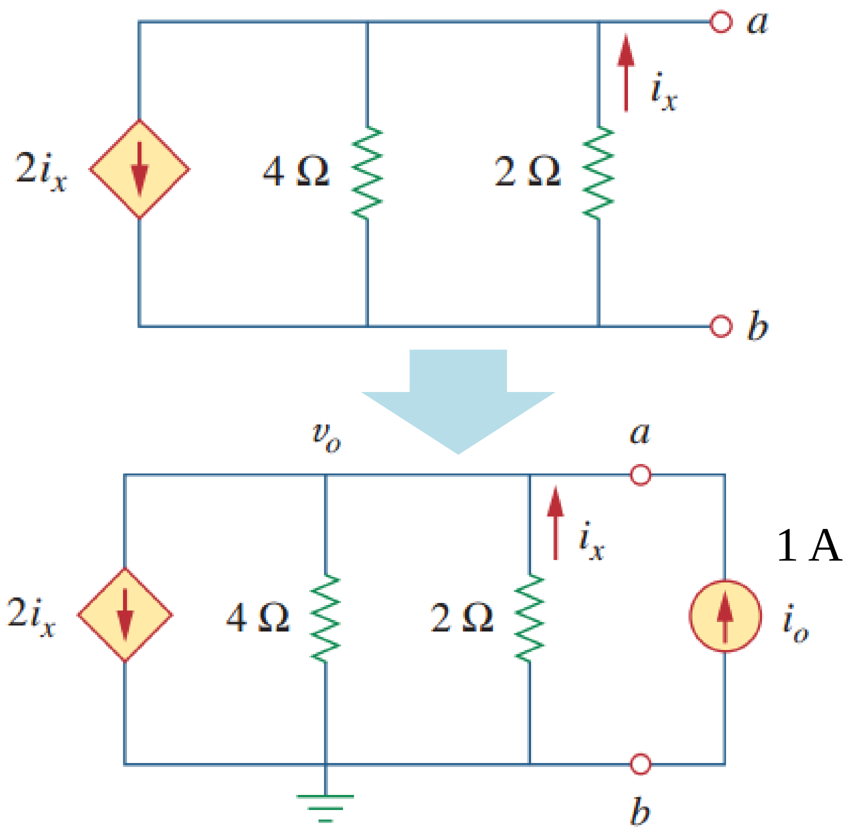
$i_x = 2$ A, $v_o = -4$ V

Therefore, $R_{Th} = v_o/i_o = -4\ \Omega$

$$v_o = R_{Th} \cdot i_o + V_{Th}$$

- Example 4.12 (When there are no independent sources)

Example 4.12: Determine the Thevenin equivalent of the circuit at terminals a-b.



a) The negative value of the resistance tells us that, according to the passive sign convention, the circuit is supplying power. Of course, the resistors cannot supply power (they absorb power); it is the dependent source that supplies the power.

b) Resistors are passive and cannot provide power, but in this circuit we do have an active device (the dependent current source). Thus, the equivalent circuit is essentially an active circuit that can supply power.

c) This is an example of how a dependent source and resistors could be used to simulate negative resistance.

d) You can use mesh analysis to check the results.

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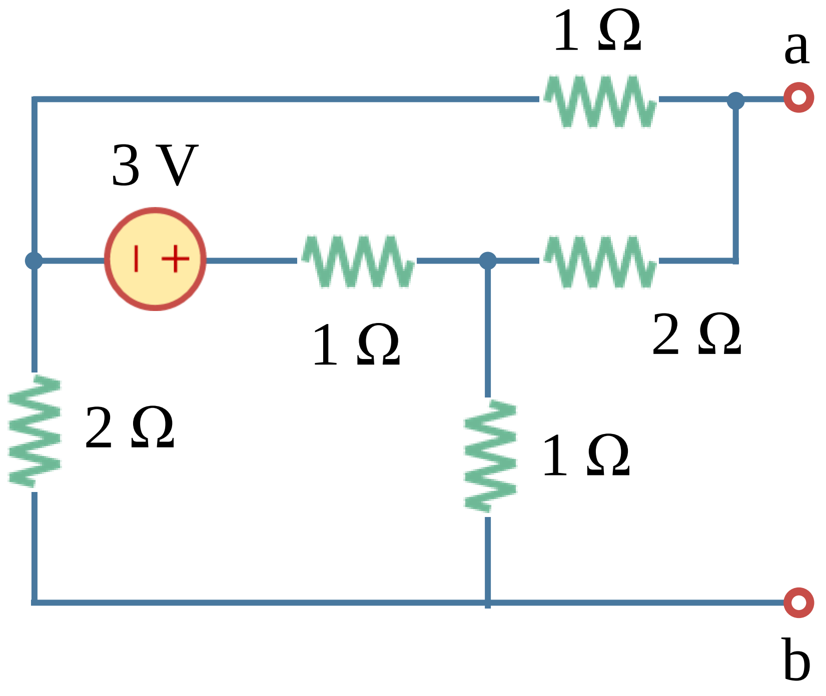
- Example 4.13

Example 4.13: Determine the Thevenin equivalent of the circuit at terminals a-b.



Solution:

a) For a complicated circuit, you can use nodal analysis to find the Thévenin equivalent directly in the form:

$$v_o = v_{Th} + i_o \cdot R_{Th}$$

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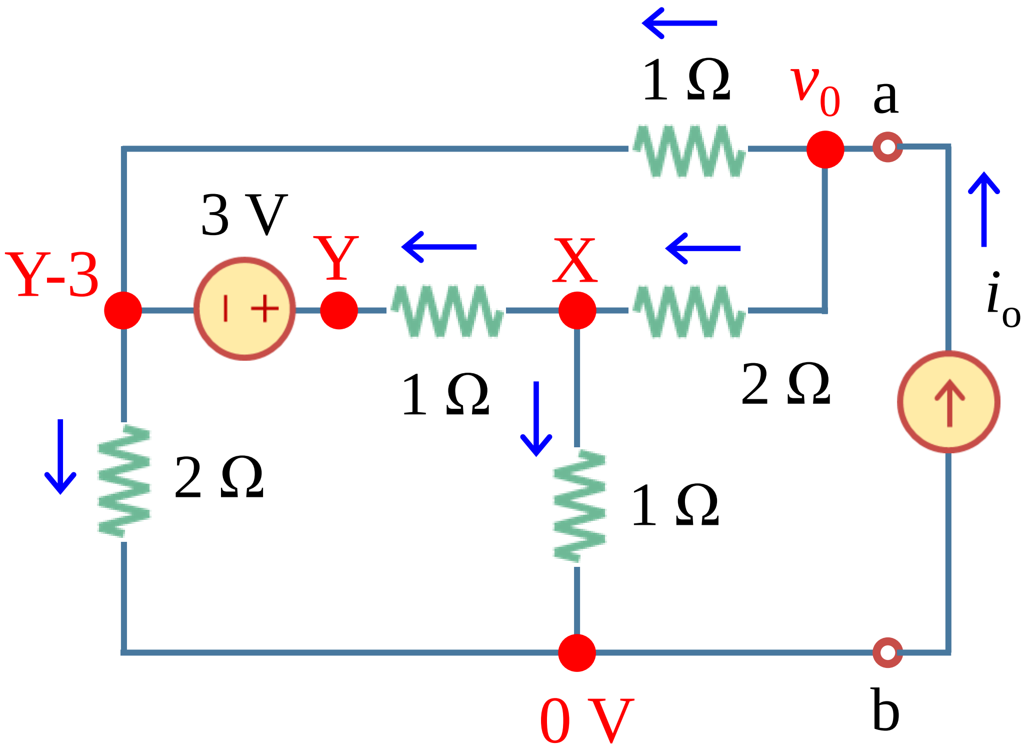
- Example 4.13

Example 4.13: Determine the Thevenin equivalent of the circuit at terminals a-b.



Solution:

b) KCL equations:

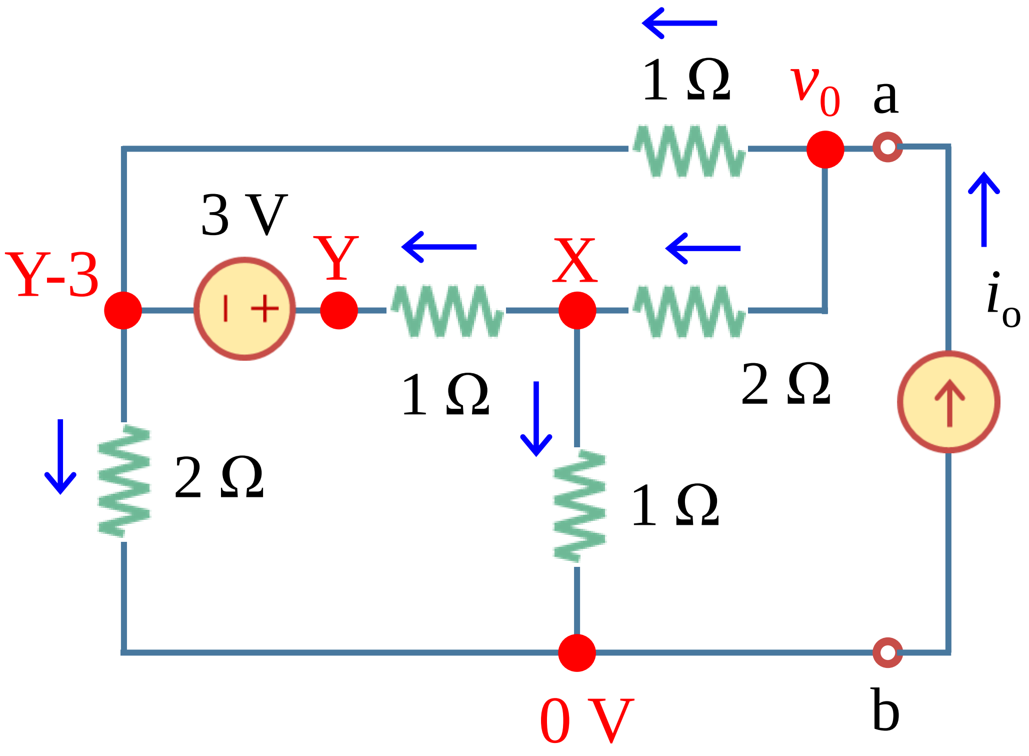
"X" node: $$\frac{v_0 - X}{2} - \frac{X}{1} - \frac{X - Y}{1} = 0$$

"Y" and "Y+3" supernode: $$\frac{v_0 - Y + 3}{1} + \frac{X - Y}{1} - \frac{Y - 3}{2} = 0$$

"v_0" node: $$i_0 - \frac{X - Y + 3}{1} - \frac{v_0 - X}{2} = 0$$

- Example 4.13

Example 4.13: Determine the Thevenin equivalent of the circuit at terminals a-b.



Solution:

c) Eliminate X and Y and solve for v_0 in terms of i_0:

$$v_o = \frac{7}{5} i_o - \frac{3}{5} = R_{Th} \cdot i_o + v_{Th}$$

Therefore $R_{Th} = 7/5\ \Omega$, $v_{Th} = -3/5$ V

Lesson 4 Circuit Theorems

4.1-4.4	Linear circuits and superposition theorem	Slides 08-31
4.5-4.6	Source transformation theorem	Slides 33-47
4.7	Thévenin’s theorem	Slides 49-60
4.8-4.9	**Norton’s theorem**	**Slides 62-70**
4.10	Maximum power transfer theorem	Slides 72-80
4.11	Summary and assignments	Slides 82-83

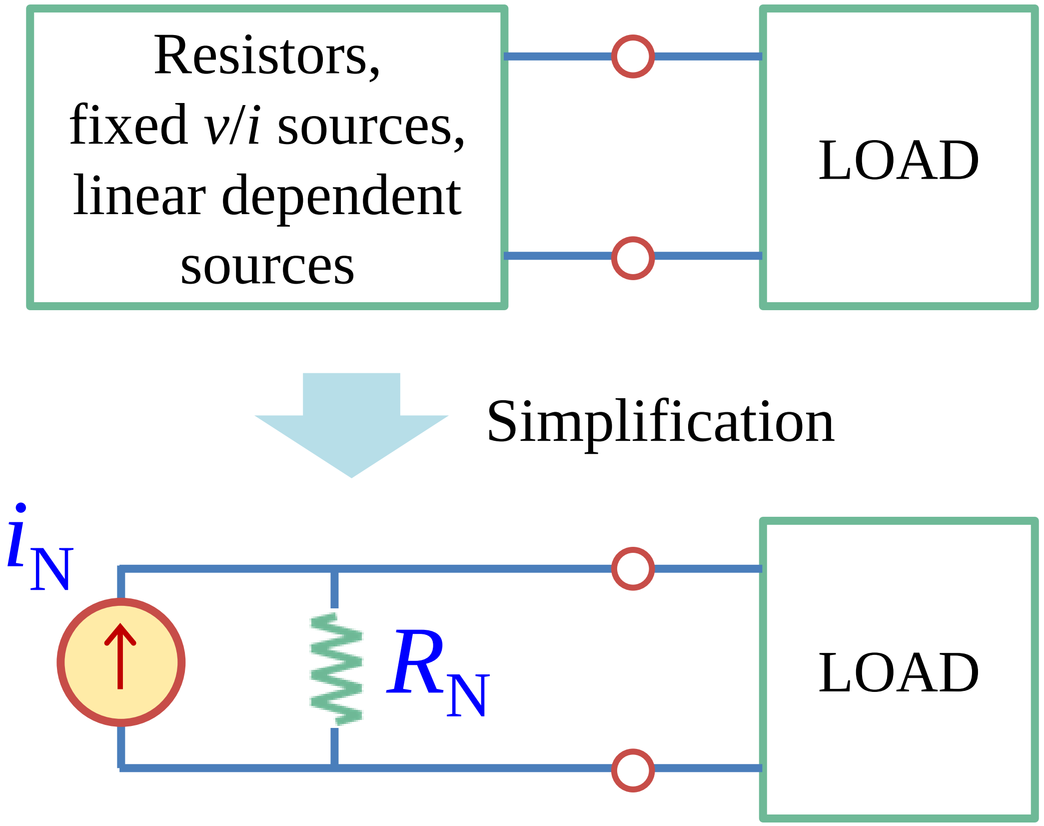
4.8 Norton's theorem (a)

- Norton's theorem (诺顿定理)

a) Norton's theorem: Any two-terminal network consisting of resistors, fixed voltage/current sources and linear dependent sources is externally equivalent to a circuit consisting of a resistor (R_N) in parallel with a fixed current source (i_N).

b) i_N is the short-circuit current at the terminals.

c) R_N is the input or the equivalent resistance measured at the terminals when all the internal independent sources are turned off.



- Norton circuit properties

a) A Norton equivalent circuit has a straight line characteristic with the equation:

$$i_o = v_o/R_N - i_N$$

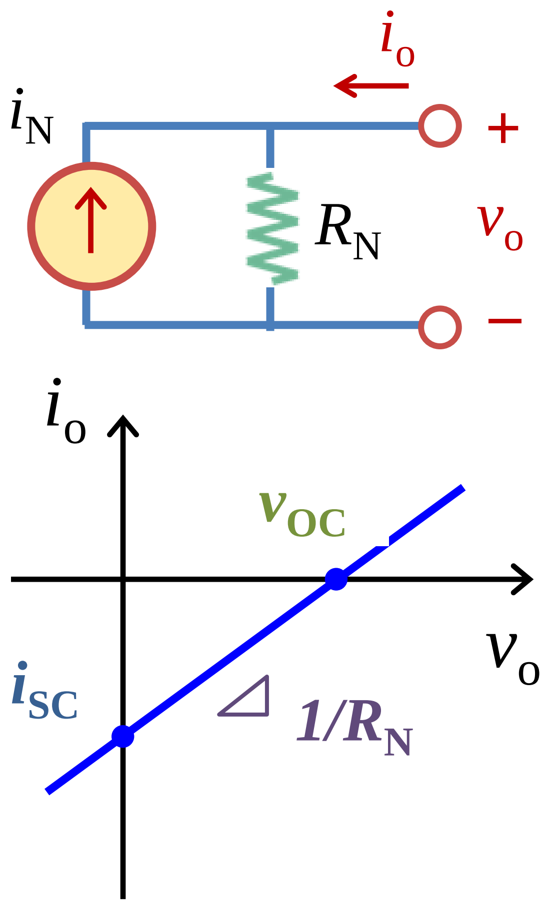
a) There are three important quantities:

Open-circuit voltage v_{OC}: If $i_o = 0$ V, $v_o = v_{OC} = i_N \cdot R_{Th}$

Short-circuit current i_{SC}: If $v_o = 0$ V, $i_o = -i_{sc} = -i_N$

Thévenin resistance v_{OC}: $R_N = v_{OC}/i_{SC} = R_{eq}$

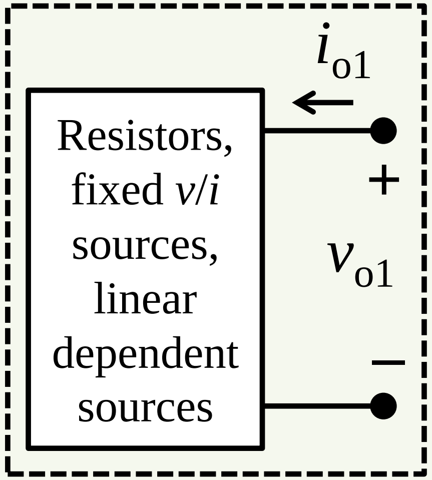
a) In any two-terminal circuit with the same characteristic, the three quantities will have the same values. So if we can determine two of them, we can work out the Norton equivalent.

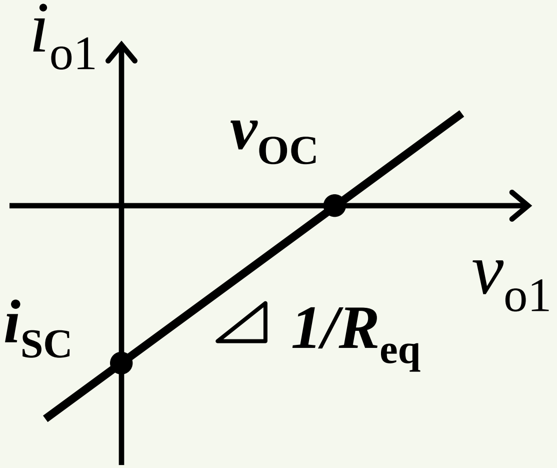


4.8 Norton's theorem (c)

- Norton's theorem

The original circuit





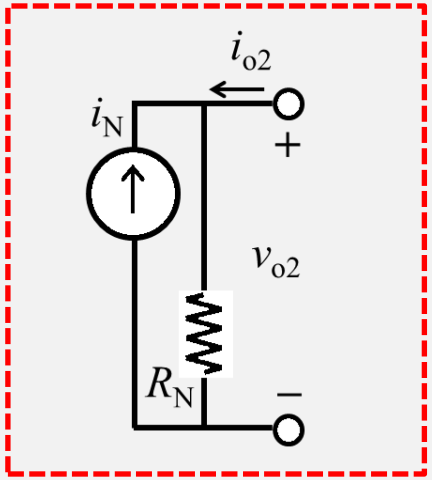
$$i_{o1} = v_{o1}R_{eq} - i_{sc}$$

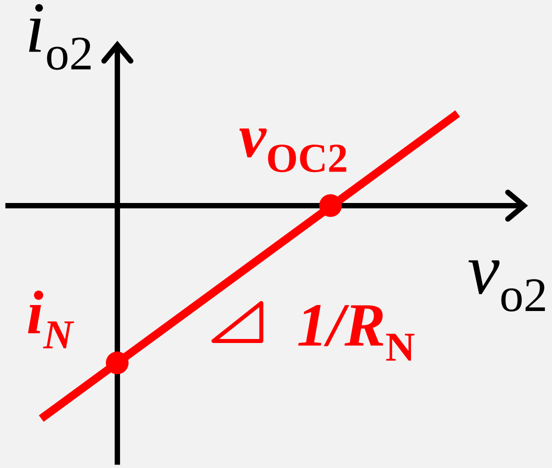
To substitute

We make them equivalent:

$$i_N = i_{SC} \ \& \ R_N = R_{eq}$$

The equivalent circuit



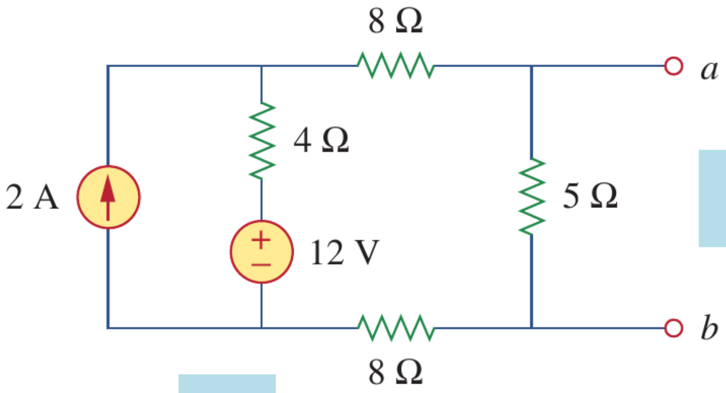


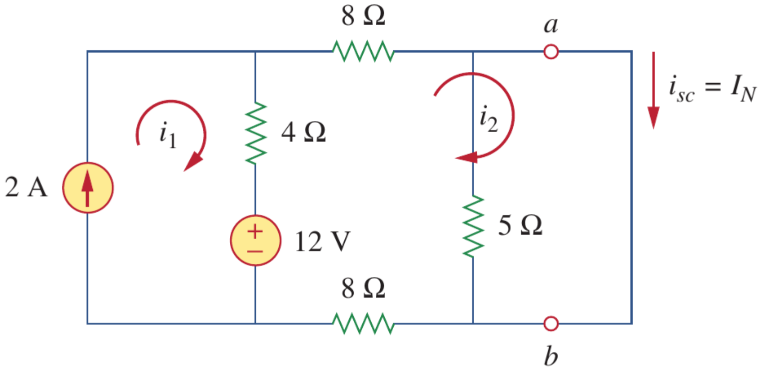
$$i_{o2} = v_{o2}R_N - i_N$$

4.8 Norton’s theorem (d)

- Example 4.14

Example 4.14: Find the Norton equivalent circuit of the following circuit at terminals *a*-*b*

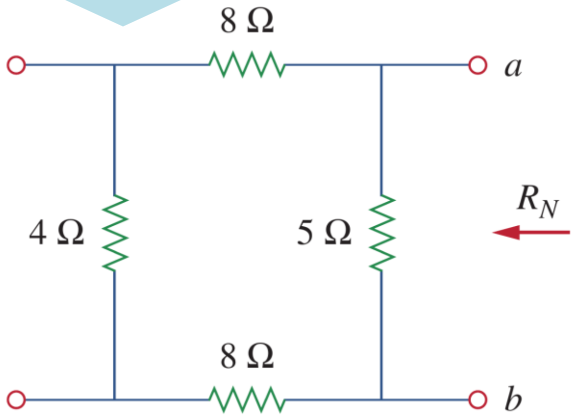




Step 1 find the short-circuit current i_{SC}

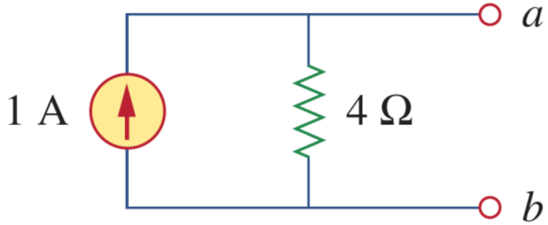
$$\begin{cases} i_1 = 2\text{ A} \\ 20i_2 - 4i_1 - 12 = 0 \end{cases}$$

So $i_{SC} = i_2 = i_N = 1\text{ A}$



Step 2 find the R_N when all independent sources are off.

$R_N = 5||(8 + 4 + 8) = 4\Omega$

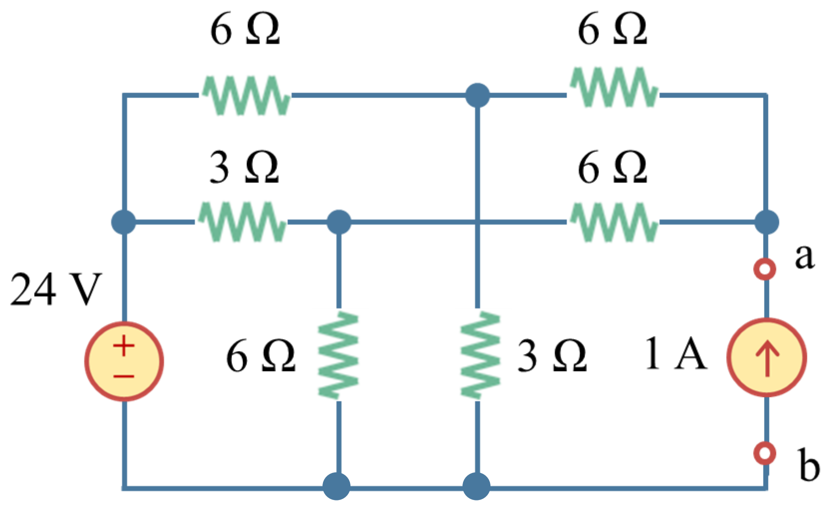


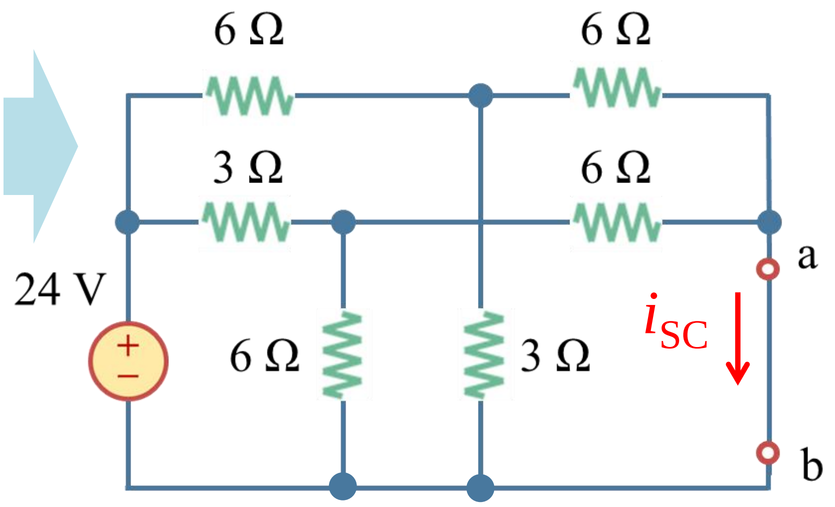
Step 3 Now you have the Norton equivalent circuit.

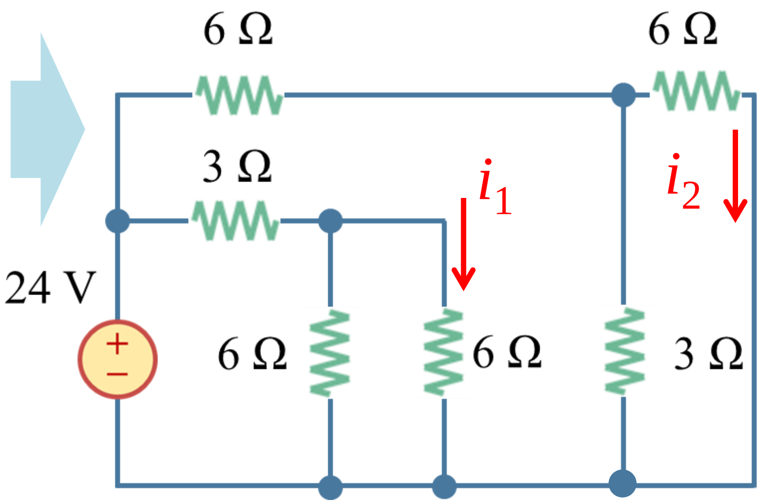
- Example 4.15

Example 4.15: Find v_{ab} in the following circuit

Solution: step 1 find the short-circuit current i_{SC}





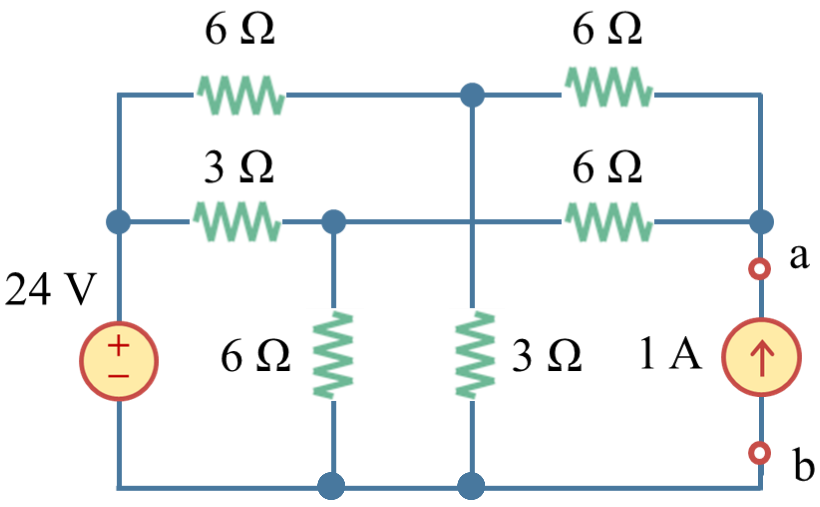


$$i_{SC} = i_1 + i_2 = \frac{24}{6 \parallel 6 + 3} \times \frac{6}{6+6} + \frac{24}{3 \parallel 6 + 6} \times \frac{3}{3+6} = 3 \text{ A}$$

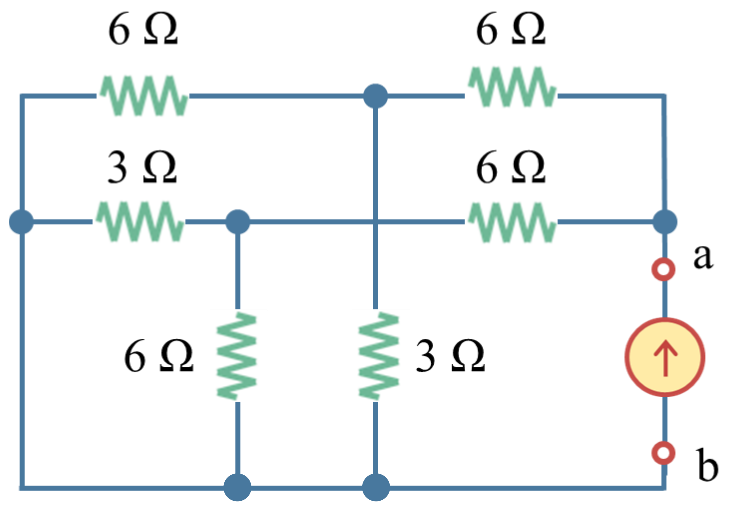
4.8 Norton's theorem (f)

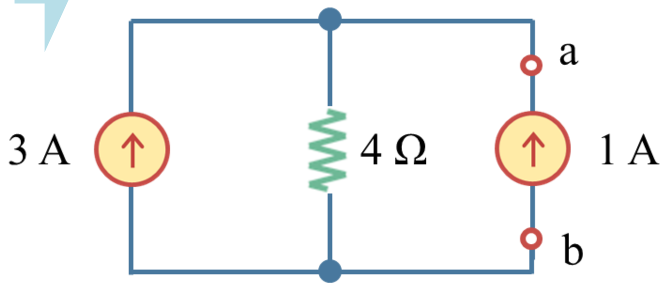
- Example 4.15

Example 4.15: Find v_{ab} in the following circuit



Solution: step 2 find the equivalent resistance R_{eq}





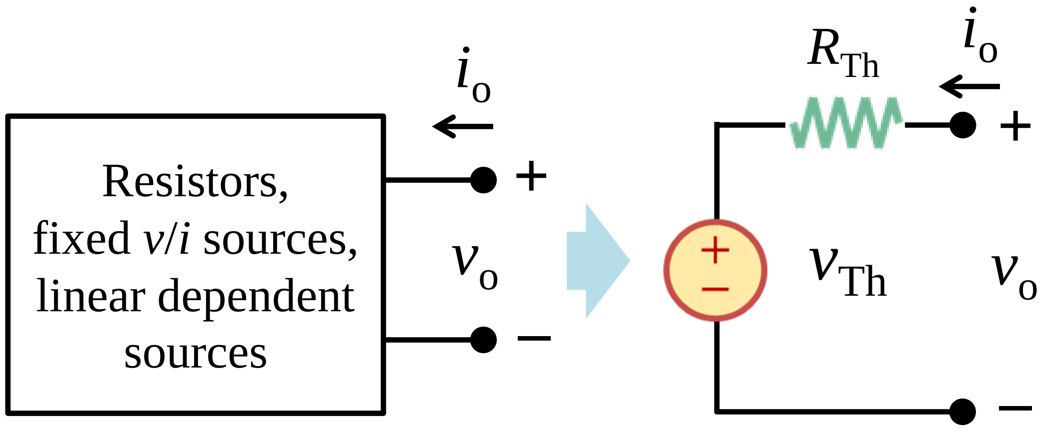
$$R_{eq} = (6||3 + 6) \,||\, (3||6 + 6) = 4\ \Omega$$

$$v_{ab} = (3+1) \times 4 = 16\ \text{V}$$

4.9 Thévenin-Norton transformation (a)

- Thévenin's theorem Vs Norton's theorem

Thévenin's theorem



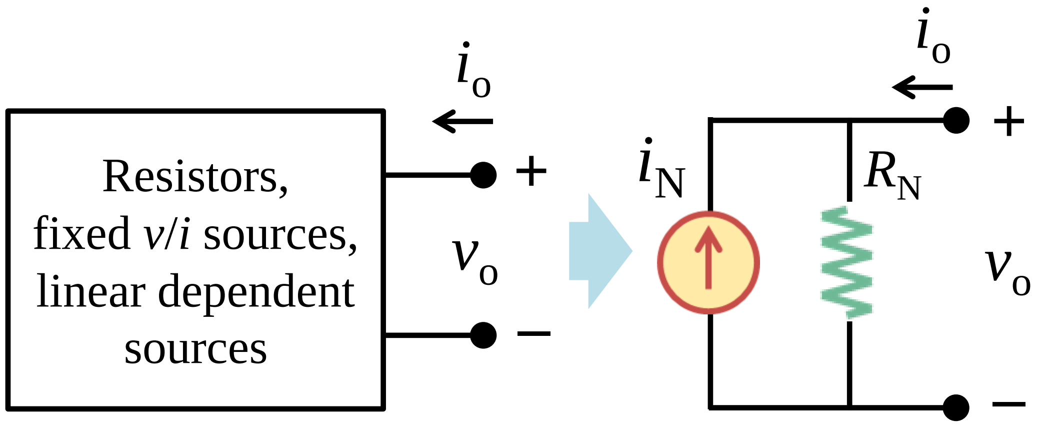
$v_o = R_{Th} \cdot i_o + V_{Th}$

Open-circuit voltage: $v_{OC} = v_{Th}$

Short-circuit current: $i_{SC} = v_{Th}/R_{Th} = -i_o$

Equivalent R: $R_{Th} = v_{OC}/i_{SC} = R_{eq}$

Norton's theorem



$i_o = v_o/R_N - i_N$

Open-circuit voltage: $v_{OC} = i_N \cdot R_N$

Short-circuit current: $i_{SC} = i_N = -i_o$

Equivalent R: $R_N = v_{OC}/i_{SC} = R_{eq}$

4.9 Thévenin-Norton transformation (b)

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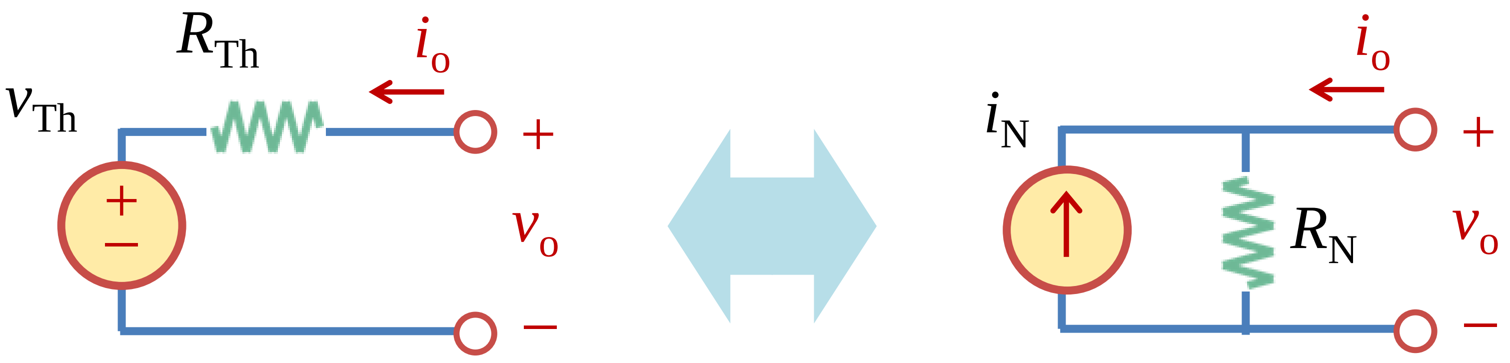
- Thevenin-Norton transformation (戴维南-诺顿变换)



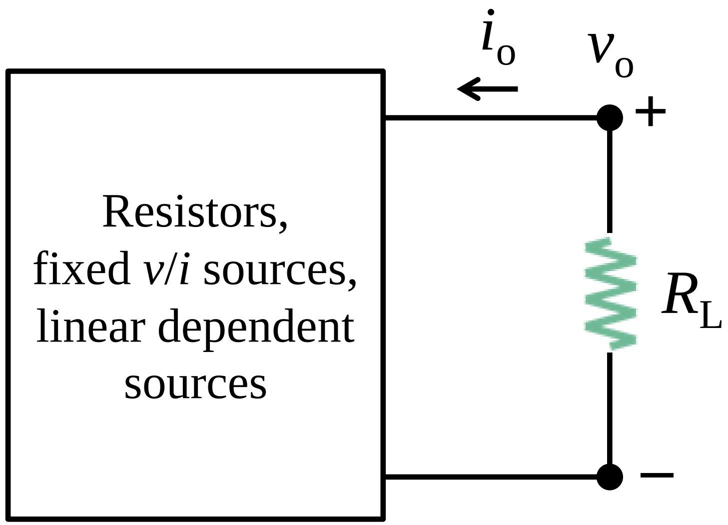
a) Thévenin's theorem and Norton's theorem are related by the Source Transformation. For this reason, Source Transform is also often called Thevenin-Norton transformation (戴维南-诺顿变换).

b) Easy to change between Norton and Thévenin: $v_{\mathrm{Th}} = i_{\mathrm{N}} \cdot R_{\mathrm{Th}}$.

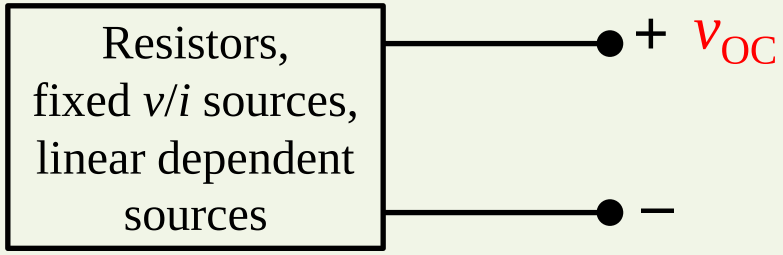
c) Usually best to use Thévenin for small R_{Th} and Norton for large R_{N} compared to the other impedances in the circuit.

4.9 Thévenin-Norton transformation (c)

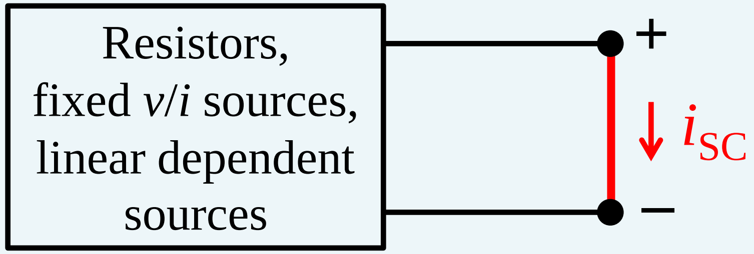
- Thévenin’s theorem & Norton’s theorem



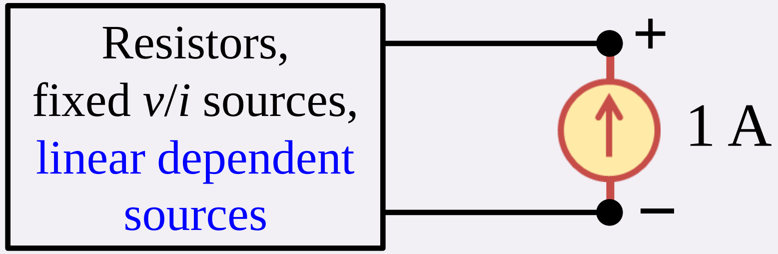
$v_{OC}/i_{SC} = R_{eq}$



To find v_{OC}, remove the load and calculate the v_{OC} using nodal/mesh analysis.



To find i_{SC}, replace the load with an ideal wire and calculate the i_{SC} using nodal/mesh analysis.



To find R_{eq}, replace the load with a 1 A current source (i_o = 1 A), turn off the independent sources, calculate the v_O, and $R_{eq} = v_O/i_O$.

Lesson 4 Circuit Theorems

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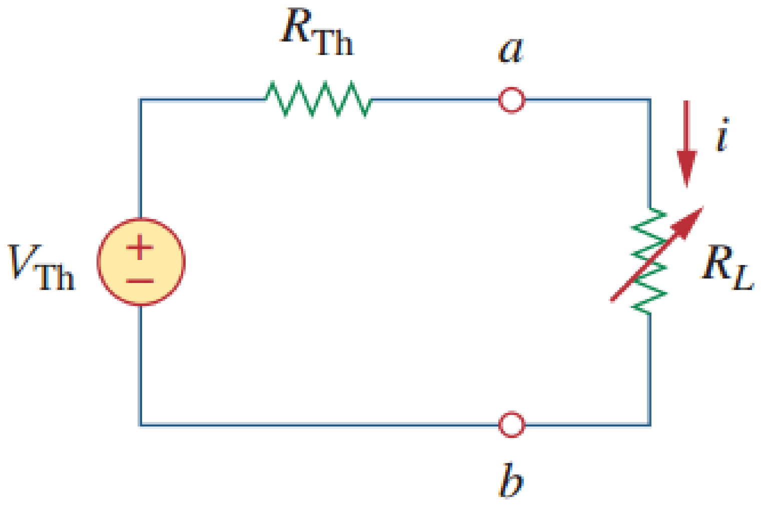
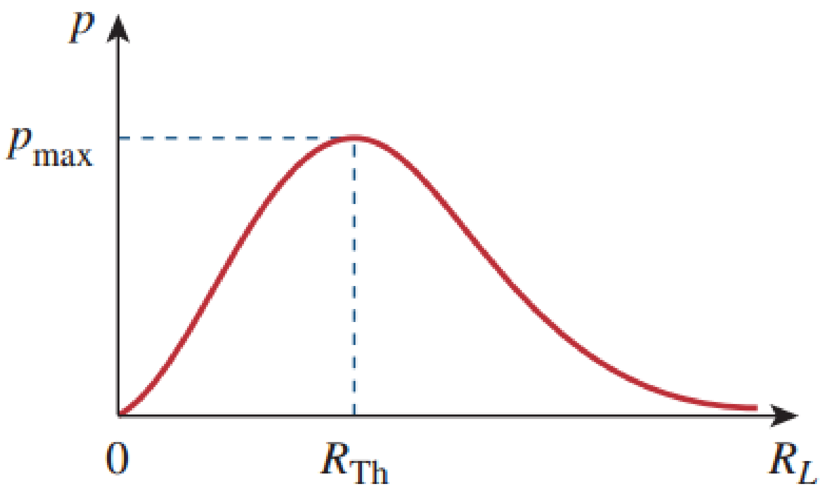
4.7	Thévenin’s theorem	Slides 49-60
4.8-4.9	Norton’s theorem	Slides 62-70
4.10	**Maximum power transfer theorem**	**Slides 72-80**
4.11	Summary and assignments	Slides 82-83

4.10 Maximum power transfer (a)

- Resistance limits power transfer



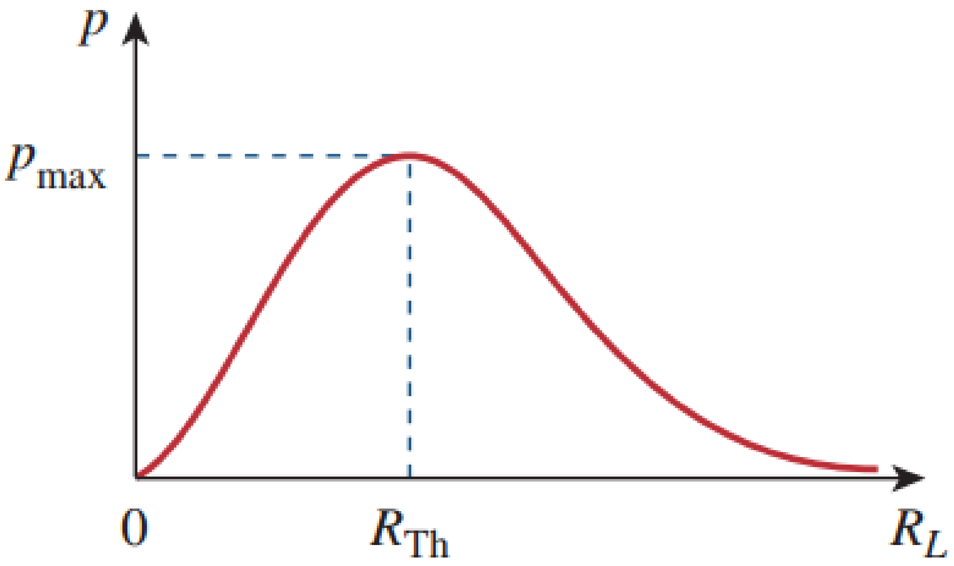
$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L}\right)^2 R_L$$



a) In many applications, the purpose of a circuit is to transfer power to a load, such as battery charging. However the internal resistance (内部电阻) of the charger puts a limit on the maximum energy transfer efficiency.

4.10 Maximum power transfer (b)

- Maximum power transfer theorem



$$R_L = R_{Th} \qquad p_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

Interesting to think: Why is it important to choose a proper charger for your cell phones?

a) Maximum power transfer theorem states that maximum power is transferred to the load when the load resistance equals to the Thevenin resistance of the circuit seen by the load ($R_L = R_{Th}$).

4.10 Maximum power transfer (c)

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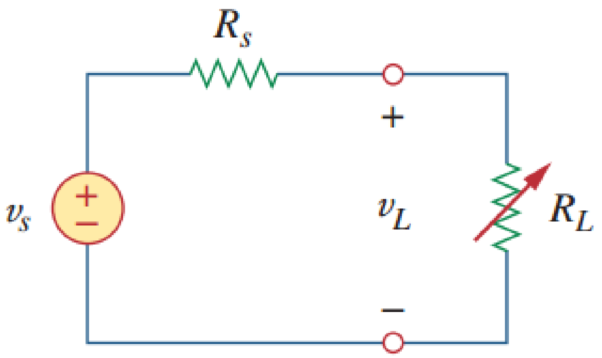
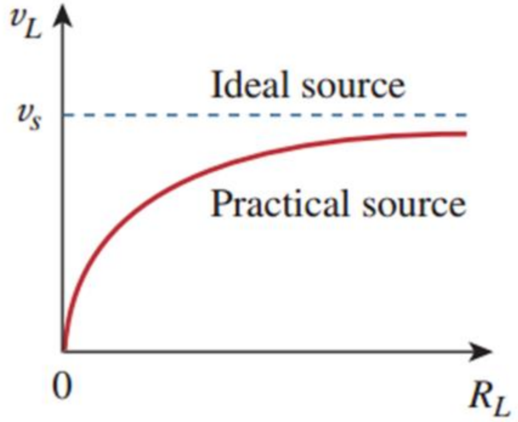
- Thevenin and Norton equivalent circuits for battery modeling

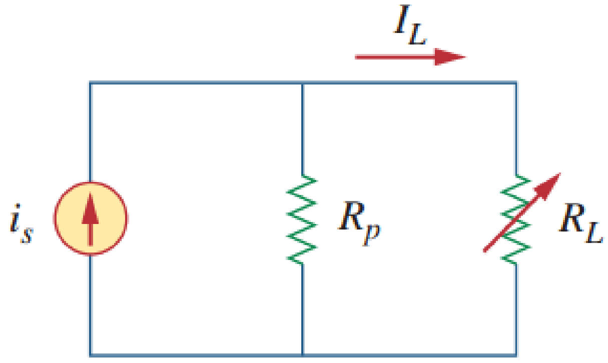
a) The Thevenin and Norton equivalent circuits are useful in modeling a real electric source, such as a battery.

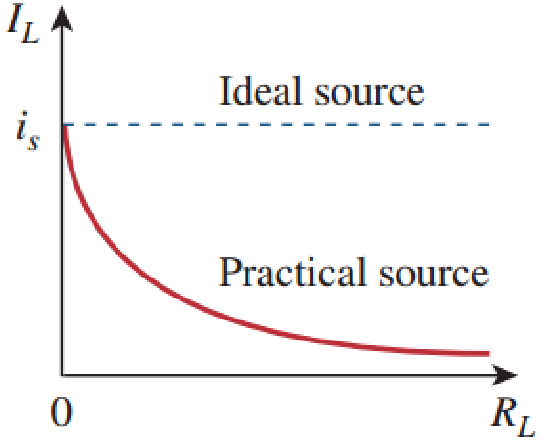
$$v_L = \frac{R_L}{R_s + R_L} v_s$$





$$i_L = \frac{R_p}{R_p + R_L} i_s$$

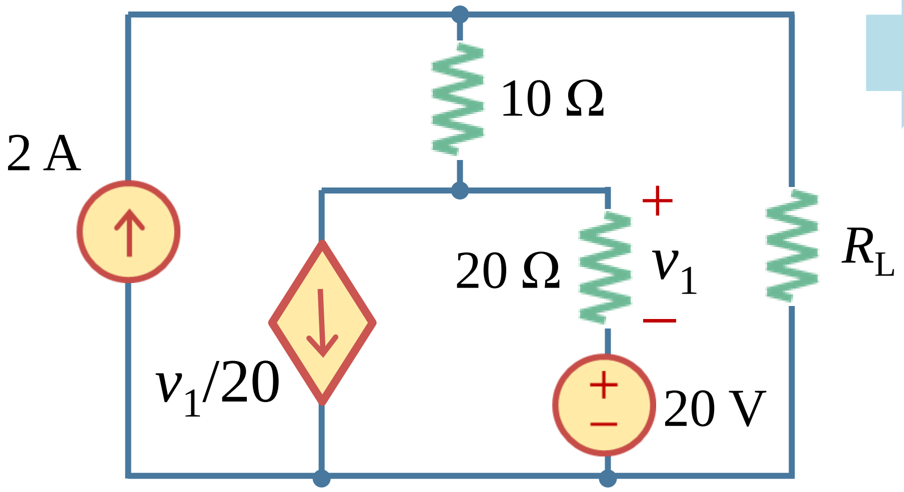




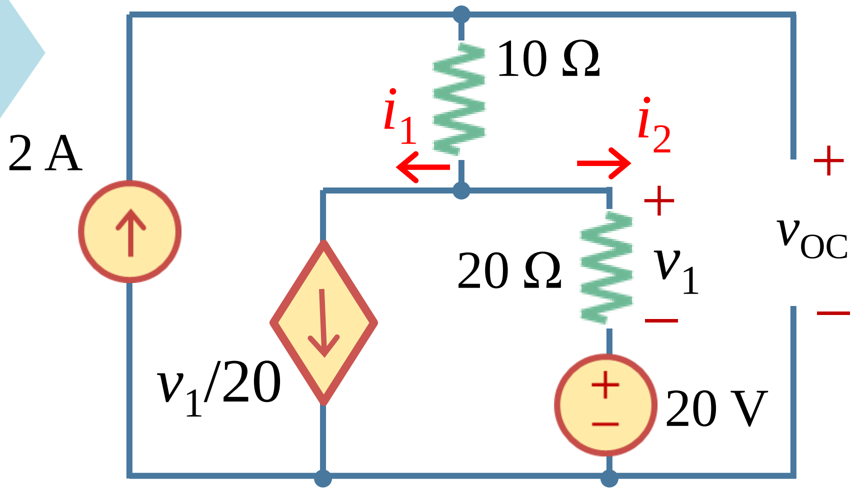
4.10 Maximum power transfer (d)

- Example 4.16

Example 4.16: Find the value of R_L to which the maximum power is transferred to R_L in the following circuit. What is the maximum power?



Solution: step 1 find the open-circuit voltage v_{OC}



$i_1 = i_2$ & $i_1 + i_2 = 2$ A

$i_1 = i_2 = 1$ A

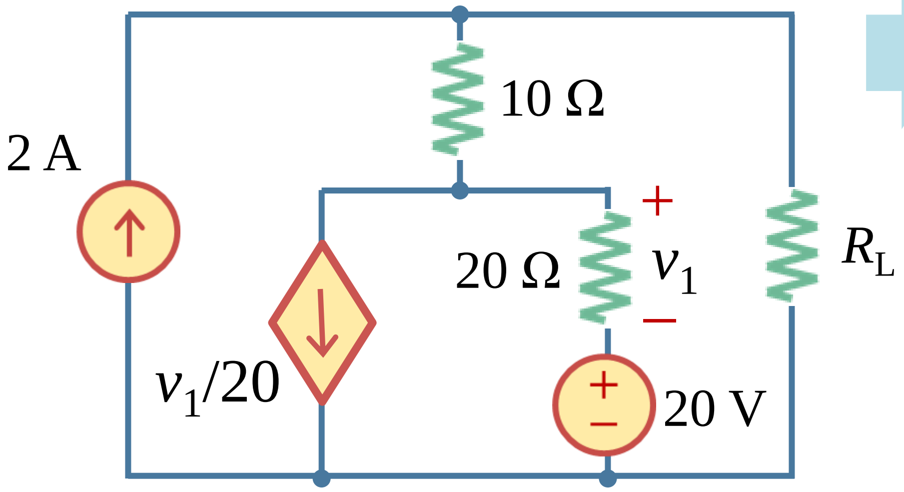
$v_{OC} = 2 \times 10 + 20i_2 + 20$

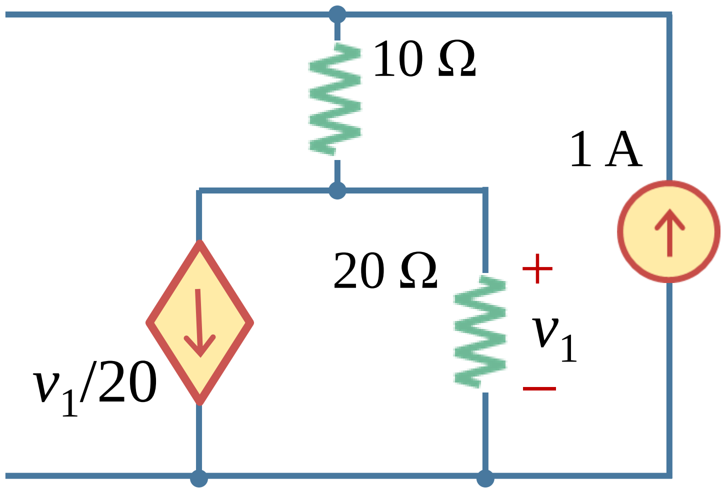
$= 60$ V

- Example 4.16

Example 4.16: Find the value of R_L to which the maximum power is transferred to R_L in the following circuit. What is the maximum power?

Solution: step 2 turn off the independent sources and find R_{eq}





$R_{eq} = 10 + 20/2 = 20\ \Omega$

So maximum power transfer is achieved When $R_L = R_{eq} = 20\ \Omega$

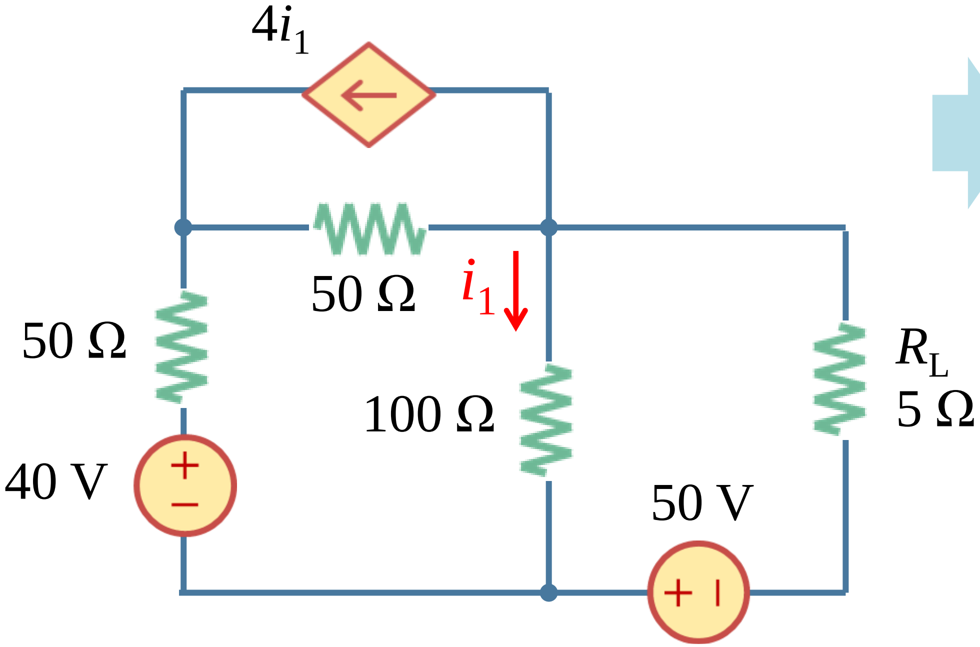
The maximum power is

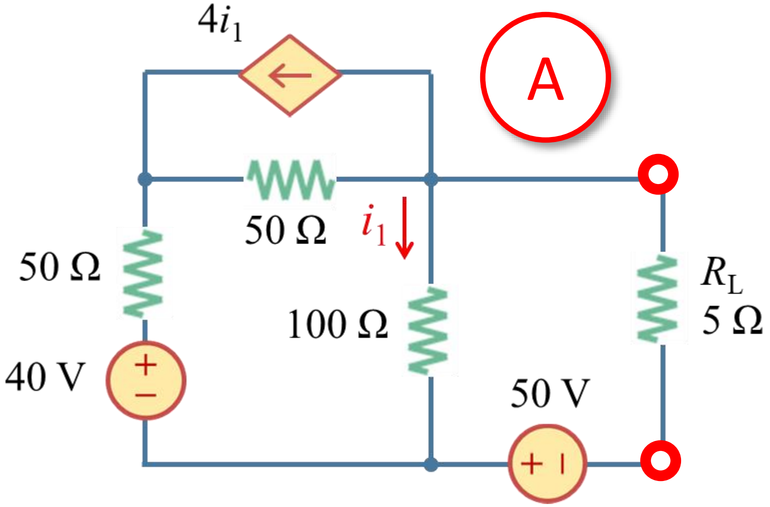
$P_{max} = 60^2/(4 \times 20) = 45\ \text{W}$

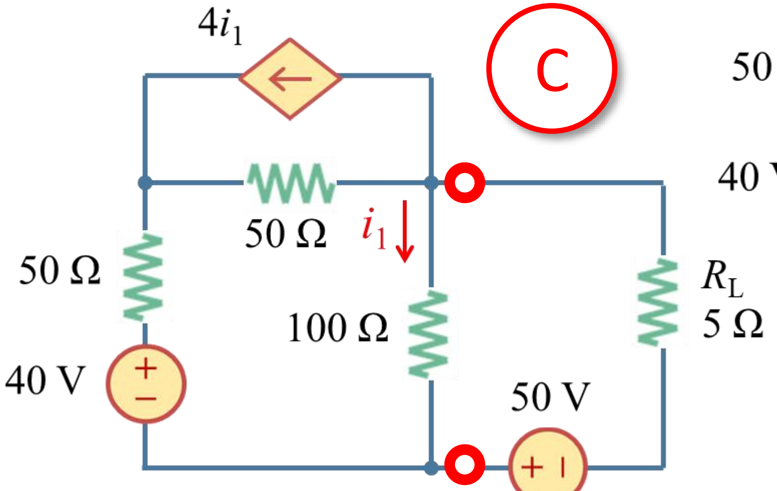
4.10 Maximum power transfer (f)

- Example 4.17

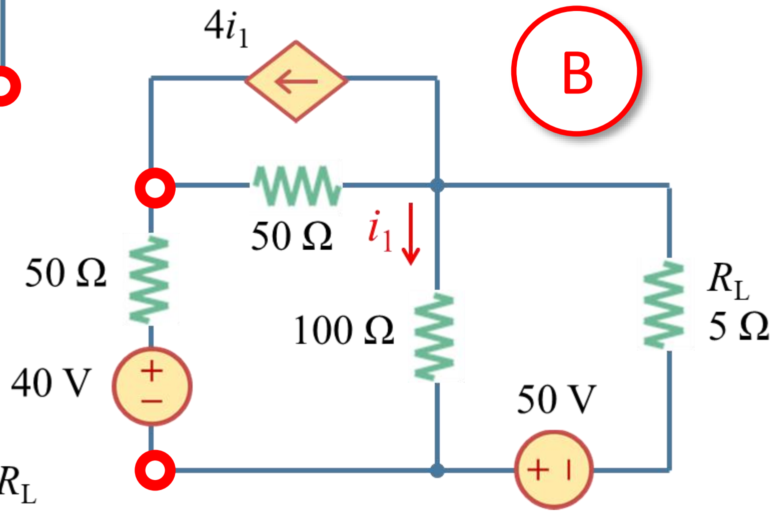
Example 4.17: Find the power that the R_L consumes. Find the R_L that the maximum power is transferred.







Which one is better, A or B or C?

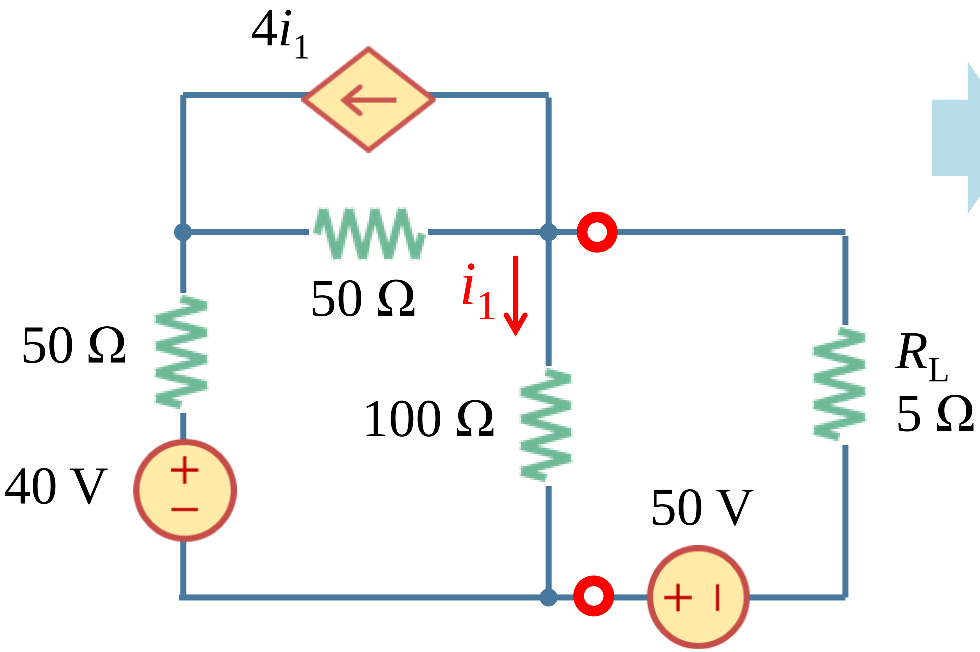


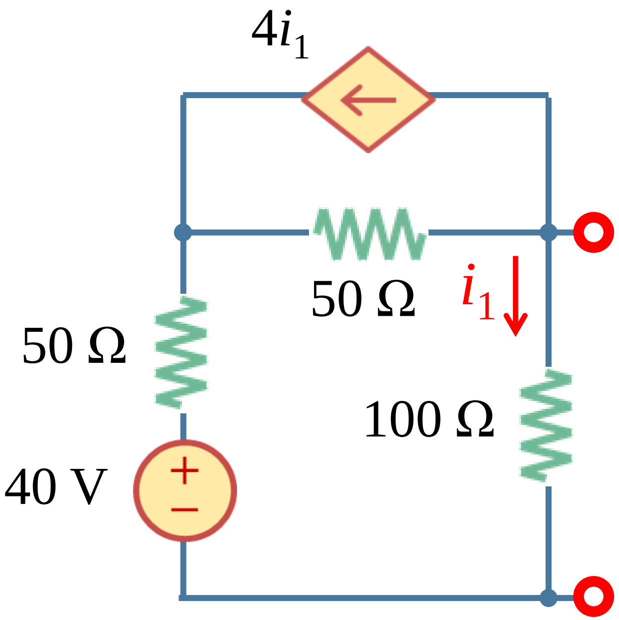
Step 1: Select proper terminals. Here C is better

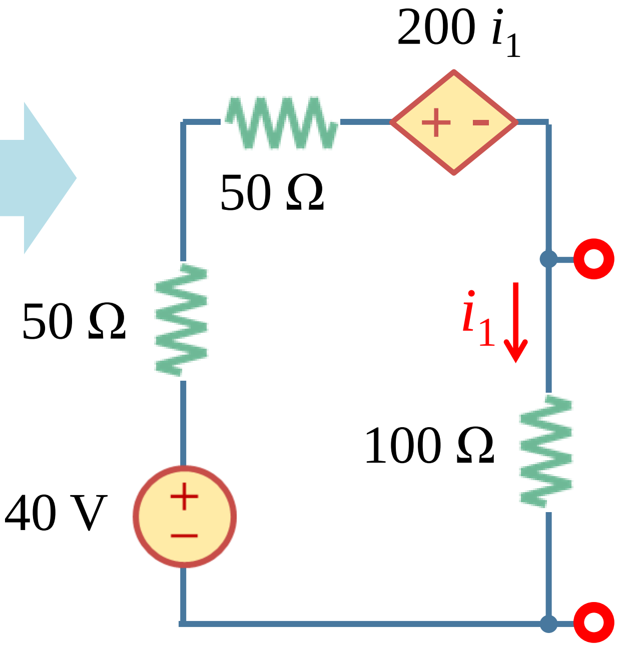
4.10 Maximum power transfer (g)

- Example 4.17

Example 4.17: Find the power that the R_L consumes. Find the R_L that the maximum power is transferred.







Solution: step 2 find the open-circuit voltage v_{OC}

$$\text{KVL: } 100\, i_1 + 200 i_1 + 100 i_1 - 40 = 0$$

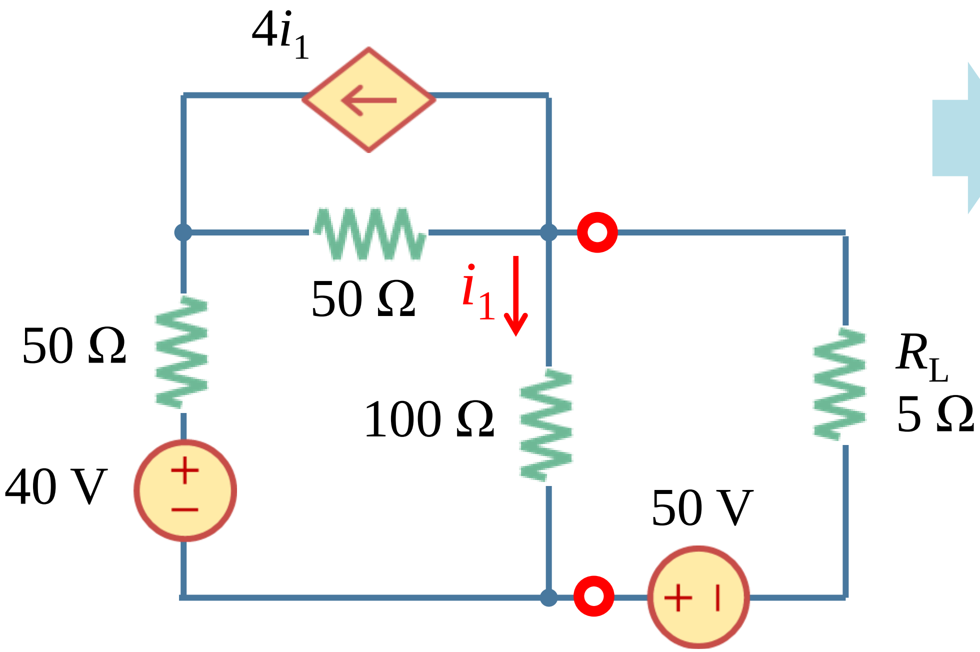
$$i_1 = 0.1 \text{ A}$$

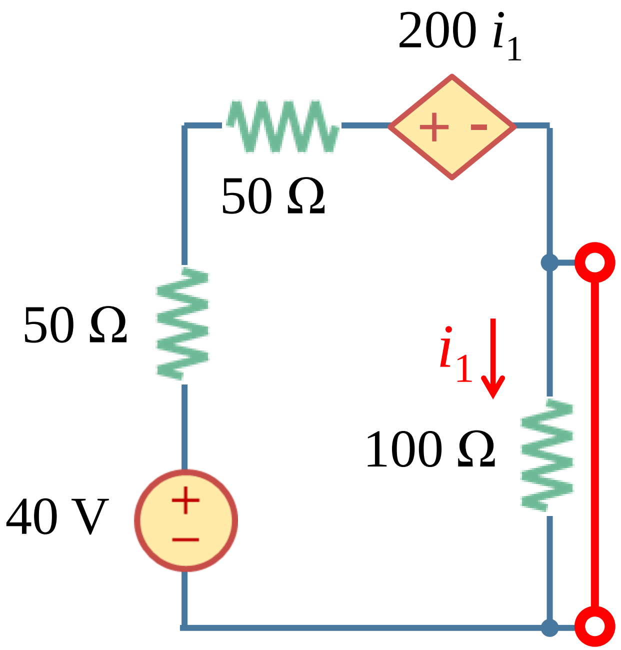
$$v_{OC} = 100 i_1 = 10 \text{ V}$$

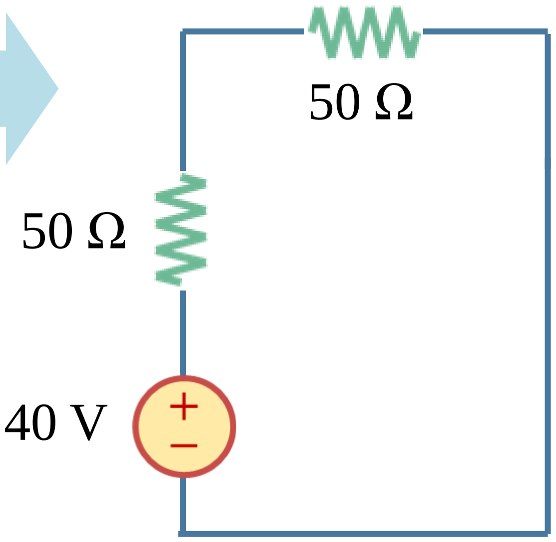
4.10 Maximum power transfer (h)

- Example 4.17

Example 4.17: Find the power that the R_L consumes. Find the R_L that the maximum power is transferred.







Solution: step 3 find i_{SC} first and then R_{eq}

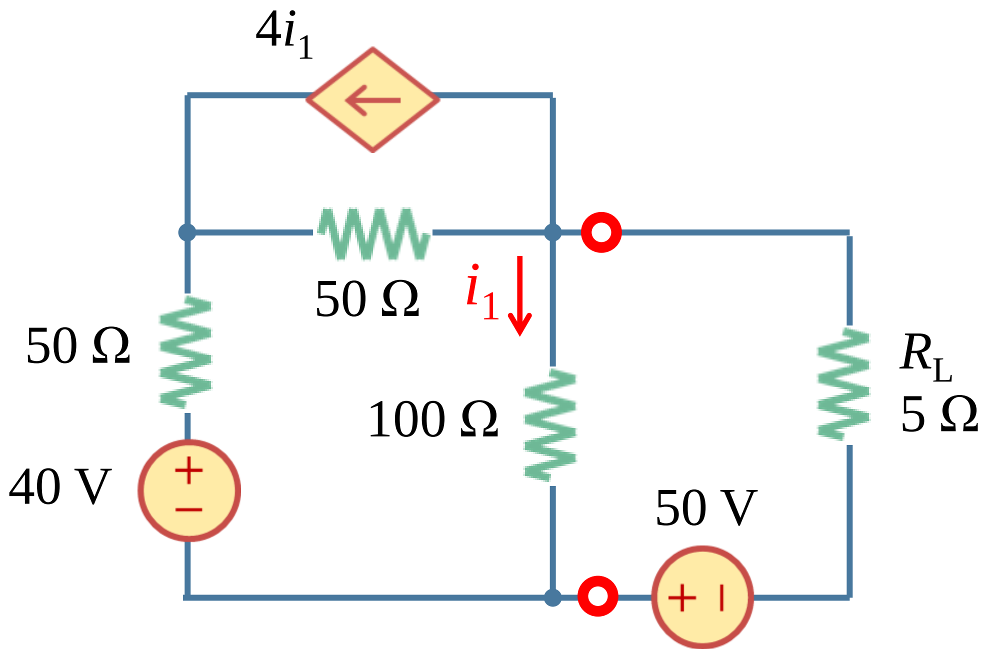
$$i_{SC} = 40/(50 + 50) = 0.4\ \text{A}$$

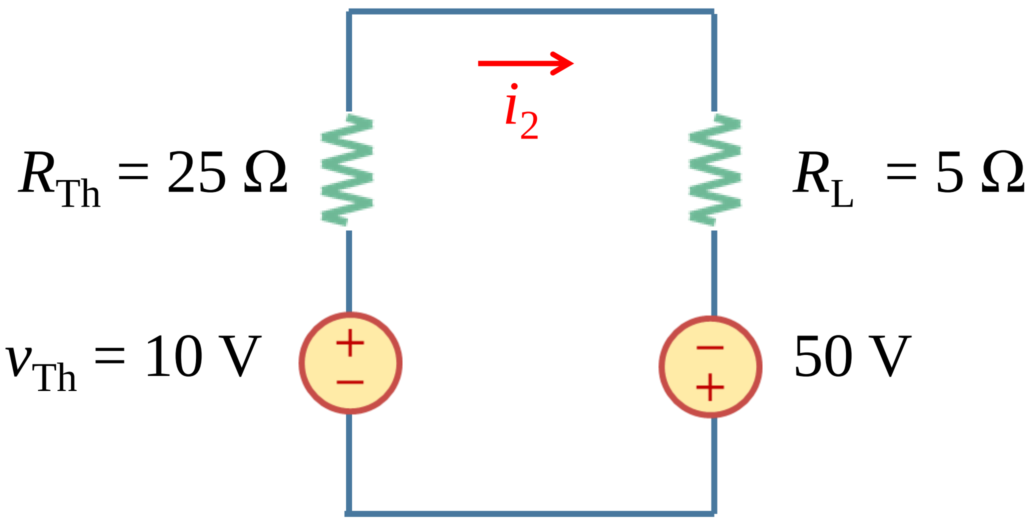
$$R_{eq} = v_{OC}/i_{SC} = 25\ \Omega$$

4.10 Maximum power transfer (h)

- Example 4.17

Example 4.17: Find the power that the R_L consumes. Find the R_L that the maximum power is transferred.





Solution: step 4 draw the equivalent and find the P

$i_2 = (10 + 50)/30 = 2$ A

$P = 5 \times 2^2 = 20$ W

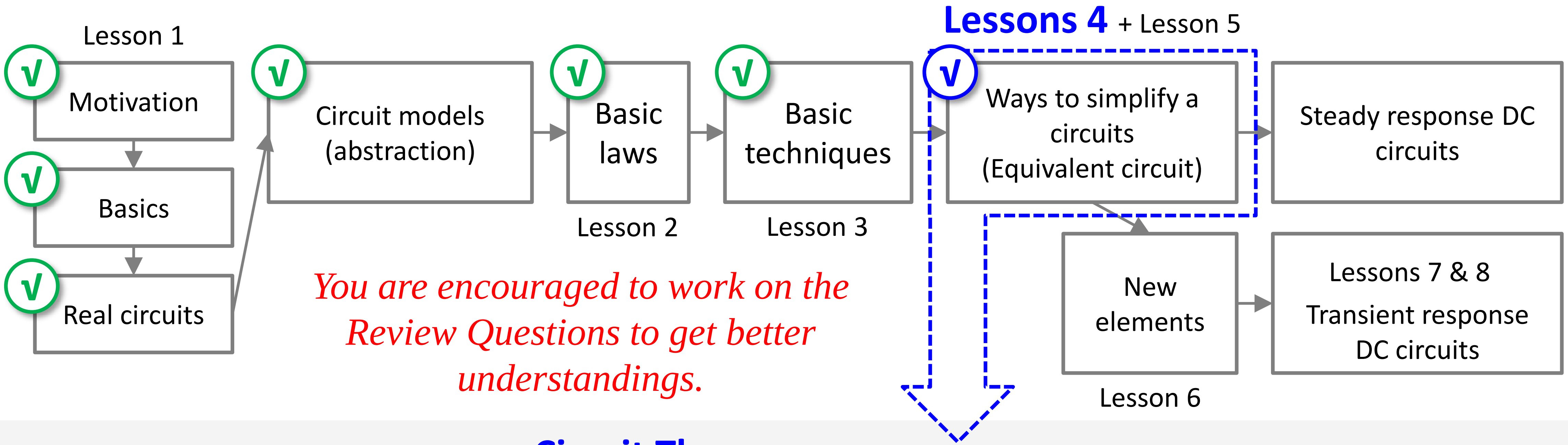
Maximum power can be transferred if $R_L = 25\ \Omega$

Lesson 4 Circuit Theorems

4.1-4.4	Linear circuits and superposition theorem	Slides 08-31
4.5-4.6	Source transformation theorem	Slides 33-47
4.7	Thévenin’s theorem	Slides 49-60
4.8-4.9	Norton’s theorem	Slides 62-70
4.10	Maximum power transfer theorem	Slides 72-80
4.11	**Summary and assignments**	**Slides 82-83**

4.11 Summary and assignments

- EE104 roadmap (DC portion)



You are encouraged to work on the Review Questions to get better understandings.

Circuit Theorems

1. Linear circuits

2. Superposition theorem

3. Source transformation theorem

4. Thévenin's theorem

5. Norton's theorem

6. Maximum power transfer theorem

4.11 Summary and assignments

南方科技大学
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- Assignments

Chapter 4:

Problems 4.14, 4.24, 4.44, 4.48, 4.54, 4.74

(a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.

(b) There are plenty of problems in the textbook, try to complete as many as you can.

Kelvin-Helmholtz clouds